

Research article

Is this AI sexist? The effects of a biased AI's anthropomorphic appearance and explainability on users' bias perceptions and trust

Tsung-Yu Hou^a, Yu-Chia Tseng^a, Chien Wen (Tina) Yuan^{b,*}^a Information Science, Cornell University, USA^b Transdisciplinary Bachelor Degree Program, D-School, National Taiwan University, Taiwan

ARTICLE INFO

Keywords:

Human-AI interaction
Explainable AI
Trust
Anthropomorphism
Bias
Gender

ABSTRACT

Biases in artificial intelligence (AI), a pressing issue in human-AI interaction, can be exacerbated by AI systems' opaqueness. This paper reports on our development of a user-centered explainable-AI approach to reducing such opaqueness, guided by the theoretical framework of anthropomorphism and the results of two 3×3 between-subjects experiments ($n = 207$ and $n = 223$). Specifically, those experiments investigated how, in a gender-biased hiring situation, three levels of AI human-likeness (low, medium, high) and three levels of richness of AI explanation (none, lean, rich) influenced users' 1) perceptions of AI bias and 2) adoption of AI's recommendations, as well as how such perceptions and adoption varied across participant characteristics such as gender and pre-existing trust in AI. We found that comprehensive explanations helped users to recognize AI bias and mitigate its influence, and that this effect was particularly pronounced among females in a scenario where females were being discriminated against. Follow-up interviews corroborated our quantitative findings. These results can usefully inform explainable AI interface design.

1. Introduction

The development of artificial intelligence (AI) has progressed rapidly in recent years, and it is now prevalent not only in everyday consumer products and applications, but also in mission-critical contexts including medical, legal, financial, and workplace decision-making that can impact people's rights and lives (Holzinger et al., 2022). For example, the *Washington Post* recently reported that AI algorithms were being used to inform human-resources (HR) decisions such as tech companies' layoffs (Verma, 2023). Unsurprisingly, these high-stakes uses of AI have raised concerns that decisions made or assisted by it could reflect biases against certain age, ethnic, gender, or socioeconomic groups, compromising people's rights to discrimination-free workplaces and livelihoods. A notable case in point was an AI tool developed by Amazon in 2014 to streamline its hiring process by reviewing and rating job applicants' resumes. This tool exhibited a significant bias against female candidates: for instance, penalizing those who had attended women's colleges. The underlying problem was later attributed to the system having been trained on a decade's worth of resumes that the company had received, predominantly from men (Verma, 2023).

The Amazon case was alarming not only because AI could have

influenced many of Amazon's hiring decisions, though the company denied this (Verma, 2023), but also because it might have represented the tip of an iceberg of gender-biased AI-assisted HR processes around the world. Assertions subsequently made by various AI-mediated recruitment companies that eliminating gender and race from their AI systems is sufficient to ensure fairer and more objective recruitment were recently criticized by Drage and Makereth (2022) as potentially misleading and overly simplistic. Chen et al. (2018) also found that, although major hiring websites including Indeed, Monster, and CareerBuilder did not rely on gender information, even by inferring it, female candidates still suffered slightly in their rankings compared to male ones. We speculated that this was because the relevant algorithms were influenced by factors weakly related to gender and/or by its gender-biased human users. Therefore, it is of pressing importance to understand how users interact with biased AI systems and the degree to which they trust them, as well as how we can mitigate their negative effects.

AI-generated results can be difficult for typical users to understand, due to the opaque nature of the algorithms, and researchers refer to this as the "black-box" problem (Collins et al., 2021). To address it, they are developing explainable AI (XAI) techniques that increase AI models'

* Corresponding author.

E-mail address: tinayuan@ntu.edu.tw (C.W.(T. Yuan).<https://doi.org/10.1016/j.ijinfomgt.2024.102775>

Received 5 April 2023; Received in revised form 28 February 2024; Accepted 1 March 2024

Available online 16 March 2024

0268-4012/© 2024 Elsevier Ltd. All rights reserved.

transparency and interpretability (e.g., Angers Schmid et al., 2022; Arrieta et al., 2020; Holzinger et al., 2022; Duan et al., 2019). User-centered approaches to XAI have also been deemed crucial to enhancing people's understanding of and trust in AI-generated outcomes. They involve providing users with clear, interpretable explanations of how an AI model works and produces its results, enabling them to make informed decisions (Angers Schmid et al., 2022; Arrieta et al., 2020; Liao et al., 2020; Duan et al., 2019). User-centered XAI research to date has included investigations of users' needs for explainability; designers' challenges in incorporating explanations into XAI products (Liao et al., 2020); how to alleviate users' cognitive load when interpreting visualized AI explanations, without compromising AI model accuracy (Herm et al., 2022); and potential means of building user trust, via interactions with explanations (Naiseh et al., 2021) or by enhancing their fairness perceptions (Angers Schmid et al., 2022).

To complement the above-mentioned work, the present study explores XAI user-interface design, with particular attention to interfaces' levels of human-like appearance and ability to offer explanations of how their systems work and their biases. For this purpose, we chose to employ a theoretical framework focused on anthropomorphism, because of that characteristic's well-documented capabilities to enhance user engagement and foster user trust (Epley et al., 2008). More specifically, the powerful social cues conveyed by non-human entities' human-like appearance and capabilities can effectively establish such entities' social presence and positively influence their users' confidence in them (Epley et al., 2008). The psychological mechanism of trust, meanwhile, helps reduce the perceived complexity of human-AI interaction and AI performance expectation (Arrieta et al., 2020; Ehsan et al., 2022; Liao & Sundar, 2022).

The prevalence of AI-enabled services and applications suggests that people are already coming into contact with them daily. However, most of these casual AI users are unlikely to understand the mechanisms that drive them, such as decision trees, machine learning, or deep learning; thus, they mainly rely on AI system interfaces for trust cues (Liao & Sundar, 2022). In these circumstances, it is timely to examine how dimensions of anthropomorphism on such interfaces serve as sources of trust for AI users. Accordingly, we designed the present research to represent the context of people being asked to make hiring decisions with the support of an AI agent that offers biased recommendations, favoring a certain gender (males in experiment 1 and females in experiment 2), and explored whether and how the agent's human-like appearance and richness of explanation about its own decisions helped our participants uncover its underlying AI's bias.

Another contribution of the present work is that it highlights the importance of distinguishing how trust arises during human-AI interaction. Although trust has been widely recognized as a pivotal outcome variable in XAI research (Arrieta et al., 2020; Ehsan et al., 2022), there have been few if any discussions of whether people's trust resides in the AI per se or in other factors such as its explanations of its decision-making. For that reason, and because AI models that follow XAI principles may nevertheless have poorly designed explainability features, this paper answers Liao and Sundar's (2022) call for researchers to consider trust in AI models and trust in communication about how those models work as separate phenomena.

The first of our two research objectives is therefore to establish how anthropomorphic cues, specifically human-like appearance and explanation richness, influence people's ability to discern when recommendations made by AI agents in simulated hiring scenarios are biased. The second is to contribute to theoretical understanding of trust in human-AI interaction by delineating whether users' trust is predominantly in the AI system itself or in the explanations provided by it, as the basis for future practical work on more effective XAI interfaces. In a 3×3 between-subjects experiment, we manipulated three different levels of human-likeness (low, medium, and high) and three types of explanations (none, information-lean, and information-rich) for decision-making in the two biased-hiring situations (experiment 1: bias against

females; experiment 2: bias against males). We found that when the AI provided information-rich explanations, its users were more able both to uncover its gender bias and to avoid it influencing their own decision-making. Nevertheless, the main locus of our participants' trust was the AI itself rather than the explanations it offered. Members of the group at a disadvantage in the scenario (i.e., females in experiment 1 and males in experiment 2) trusted biased AI less than their advantaged counterparts did, and were also influenced less by it. However, these effects were only salient when the biases were better recognized by society in our given context, such as in experiment 1 (bias against female applicants), compared to less common biases such as in experiment 2 (bias against male applicants).

We also investigated this problem space qualitatively, and those results contribute to information-management studies by unraveling the intricate dynamics of human-AI interaction in high-stakes decision-making scenarios, and provide valuable insights for the design and implementation of AI systems in information-rich environments. Lastly, our work contributes to the XAI literature on user-centered system designs in the following ways:

- Use of a high-stakes task – hiring decision-making – to examine how users respond to AI's gender bias based on its appearance and explanations;
- Differentiation among levels of two anthropomorphic cues that can be applied to XAI system interface designs, with the wider aim of helping users identify and avoid adopting AI biases;
- Investigation of users' responses to AI's gender bias through two parallel experiments – one featuring bias against males and the other, bias against females – and a follow-up interview study; and
- Separation of the intricate mechanisms of trust into two types: trust in AI and trust in its explanations for its decision-making.

In the subsequent sections, we will comprehensively explore user responses to AI gender bias in high-stakes hiring decisions. First, the Literature Review section establishes the scholarly and real-world contexts for our research, i.e., the evolving landscape of AI biases and the necessity for developing and improving XAI. Then, the Research Design and Methods sections set forth our research design, encompassing two experiments and a qualitative interview, and our rationale for selecting it. Next, the Results section showcases the outcomes of our quantitative and qualitative analyses; and the Discussion section highlights some of those outcomes' theoretical and practical implications, notably regarding the pivotal role of anthropomorphism and trust in shaping AI users' perceptions. Lastly, the Conclusion summarizes our key findings and outlines avenues for future research.

2. Literature review

2.1. Artificial intelligence and algorithmic bias

Despite AI companies' claims that their products offer superior fairness (Raghavan et al., 2020), automated decision systems can harbor algorithmic biases (Rovatsos et al., 2019). Such biases have been traced to varied origins, including but not limited to data collection and modeling processes (Mehrabi et al., 2021). Our study adopts Friedman and Nissenbaum's (1996) definition of *bias* as systematic and unfair discrimination, often tied to characteristics like race, gender, religion, or nationality (see also Akter et al., 2021; Kordzadeh & Ghasemaghahi, 2022; Varsha, 2023; Yuan et al., 2023).

Biases in computerized systems were recognized as early as 1988, when St. George's Hospital Medical School in London was found to employ programs that discriminated against females and people with non-European names when assisting with student-admissions decisions (Lowry & Macpherson, 1988). In recent years, these issues have become more prominent due to the rise of AI technologies including machine learning (ML) and natural-language processing (NLP), which depend on

extensive data sets and are thus vulnerable to historical biases (Danks & London, 2017; Mehrabi et al., 2021; Roselli et al., 2019). Bolukbasi et al. (2016), for instance, revealed how gender stereotypes arose in an NLP framework trained on Google News articles, highlighting that even content created by professional journalists is not immune to such biases.

Biased AI systems can damage society by exacerbating inequality and injustice (Varsha, 2023), notably through unfair decision-making in critical contexts like healthcare (Gaonkar et al., 2020), hiring (Raghavan et al., 2020), finance (Heaven, 2022), and the law (Werpachowska, 2020). This problem is compounded by its potential to become a self-reinforcing loop, in which biased AI outcomes are adopted by humans, leading to biased decisions which then become inputs for the AI's subsequent iterations (Iwasiński, 2020).

Due to this broad impact of AI bias, numerous scholars, research institutions and companies have proposed ways of mitigating it (e.g., Varsha, 2023; Whittaker et al., 2019). However, research also highlights that AI bias-related studies in the management domain are still in their nascent stages and require further empirical investigations at the interaction level: i.e., how people react to biased AI and whether and how AI biases influence human decision-making (Varsha, 2023). The present paper is in part an effort to bridge that gap.

2.2. Bias perception and decision-making

People's interactions with biased AI can be said to take place on two levels: cognitive and behavioral. On the first, people may or may not perceive AI as biased. On the second, their decision-making may or may not be influenced by the biased AI; and where it is, they may or may not realize that this has happened.

Cognition. Research focused on the cognitive level has explored people's perceptions of biased AI, often in combination with the broader concept of perceived fairness (e.g., Angers Schmid et al., 2022). Wang et al. (2020), for instance, examined how three factors – algorithms' outcomes, development and usage, and user characteristics – influenced people's perceptions of algorithmic fairness. Their participants generally considered an algorithm fairer if its outcomes were favorable to them, though this effect was moderated by factors such as gender. The same authors also explored whether belonging to a group benefiting either from bias or from algorithm transparency influenced people's fairness perceptions. Although the results of that exploration were non-significant, we believe that they merit further investigation for two reasons. First, Wang et al.'s experimental context – hit approval on Amazon Mechanical Turk (MTurk) – typically does not hinge on gender, suggesting a need to reevaluate their results in contexts with stronger demographic-bias risks, such as workplaces (Heilman, 2012). And second, their transparency manipulation focused solely on indications of algorithms' openness to the public. Given the growing importance of AI explainability, we advocate including this critical aspect of transparency in AI-bias research.

Another cognitive factor that affects people's perceptions of AI bias is their own awareness of stereotypes and discrimination in society, i.e., their stigma consciousness (Pinel, 1999). Jussupow et al. (2021), for example, found that the more AI users were aware of gender stereotypes, the more likely they were to perceive a biased AI as untrustworthy. On the other hand, users who were less aware of such social stereotypes sometimes judged an AI that explicitly expressed its gender bias as more trustworthy, simply because it was more transparent.

Behavior. Fewer studies have looked beyond perceptions and attitudes to actual behavior, i.e., whether and how people's decision-making is influenced by a biased AI. In one, Ebrahimi and Hassanein (2019) investigated whether increased data transparency would influence people's reactions to a biased AI system's discriminatory recommendations. Specifically, after providing information such as demographic transparency charts, the researchers expected to observe not only increases in perceptions of the AI's negative effects but also, behaviorally, a decreased level of acceptance of the AI's suggestions.

Our study differs from such prior work in that it looks at 1) how biases may or may not be transferred through multiple rounds of interaction, and 2) whether people's acceptance of an AI system is affected by its appearance and explainability, as discussed in the following sections.

2.3. The theoretical framework of anthropomorphism and XAI

Recent XAI research underscores the importance of transparency and explainability to fostering trust in AI and encouraging its use (Angers Schmid et al., 2022; Sokol, 2019). Explainability has multiple dimensions, including transparency, auditability, user responsiveness, accessibility, trustworthiness, and ethics (Arrieta et al., 2020; Liao et al., 2020; Liao & Sundar, 2022). The present study specifically focuses on XAI's transparency dimension, which entails providing users with readily understandable information about systems' capabilities, limitations, data sources, and result derivation (Arrieta et al., 2020; Liao & Sundar, 2022). From our user-centered perspective, it is critical for research of this kind to facilitate transparency designs that enable users to make informed, unbiased decisions when acting upon AI output.

For human-centered designers, visibility is a crucial principle for improving human-computer interaction (Norman, 1999). Nevertheless, when it comes to AI-enabled tools and applications, visibility into how algorithms work is usually lacking. While extensive academic research has focused on improving AI explainability via computational solutions that reveal AI learning processes and model decision sequences (e.g., Ribeiro et al., 2016), scholars have also emphasized the importance of social dimensions and end users' perspectives (Liao et al., 2020; Miller, 2019b). Mathews (2019), for instance, advocated that layperson-friendly explanations be incorporated into AI-enabled applications. It has also been argued that user-centered XAI could both allay widespread concerns about over-reliance on AI at the expense of human agency (Arrieta et al., 2020) and help its users adapt as a means of overcoming unexpected problems (Adadi & Berrada, 2018; Liao & Sundar, 2022).

Grounded in the theoretical framework of anthropomorphism, the present work proposes that employing interface designs with human-like appearances and capabilities will enhance XAI transparency and thus foster user trust. Extensive research on human-AI interaction has explored how AI provides human-like cues through behaviors and forms (Letheren et al., 2021; Miao et al., 2021; Santoro & Mornin, 2023). Visible and tangible cues in AI can increase user engagement by exhibiting behavioral and form realism, distinguishing systems from their disembodied counterparts (Letheren et al., 2021; Nowak & Biocca, 2003; Short et al., 1976). It has also been shown that different types of human-like cues have potentially divergent effects, and thus, that each warrants separate examination (Hu et al., 2021). We extend these previous efforts by proposing and testing the idea that system interfaces' mimicry of human appearance and behaviors offers perceptual and action cues whereby their users can come to understand AI algorithms.

Additionally, the capabilities an AI displays may assist with model transparency by addressing the opacity of machine-/deep-learning algorithms and fostering user trust in AI interactions (Liao & Sundar, 2022). In contrast to prior studies, our approach, grounded in the anthropomorphism framework, contextualizes individual perceptions of AI within biased situations and investigates how explainable, human-like cues from AI impact its users' behavior, in contrast to prior studies that have primarily focused on AI's own attributes (e.g., Jackson et al., 2020; Santoro & Mornin, 2023).

The main concepts and identified issues of this study are summarized in Table 1.

3. Theoretical Background and Hypotheses Development

3.1. Anthropomorphic Levels: Appearance and Capability to Explain

The following two subsections respectively elaborate on the two key

Table 1

Key Concepts and Key Issues of the Current Study.

Construct	Definition	Key issues and identified gaps	Details	References
Bias	Systematic and unfair discrimination in various forms, often tied to characteristics like race, gender, religion, or nationality.	Concerns about bias in AI tools that may evolve into sociotechnical ones.	Bias in AI can result from different levels: data to algorithm, algorithm to user, and user to data (i.e., datasets, mathematical models, users influenced by historical or social contexts.) Evaluating the extent to which individuals are influenced by biased AI applications, particularly within crucial domains such as hiring decisions.	Friedman & Nissenbaum (1996), Mehrabi et al. (2021), Akter et al. (2023)
Anthropomorphism	Imbuing the imagined or real behavior of non-human agents with human-like characteristics, motivations, intentions, and emotions	1. Lack of understanding regarding how anthropomorphism influences users' perception of AI biases. 2. Limited investigation into the relationship between anthropomorphism and trust in AI recommendations. 3. Insufficient exploration of the impact of human-like appearance and capabilities on user trust in AI systems.	Examining if users perceive AI biases differently based on anthropomorphism levels, including human-like appearance and capabilities in AI interfaces, and whether higher anthropomorphism boosts trust in AI recommendations.	Epley et al. (2007)
Human-likeness (anthropomorphism from appearance)	The extent to which an artificial intelligence (AI) agent's shape appears human			Miao et al. (2022)
Explainability (anthropomorphism from capabilities)	The extent to which an AI system can make itself transparent and understandable			Liao et al. (2020)
Information richness	The degrees of capacity in communication, influenced by four factors: language variety, cue multiplicity, feedback immediacy, and personalization in a given piece of information	What the role of information richness and the level of such richness are in enhancing explainability in AI systems.	Examining information richness levels, such as the depth and detail of explanations provided by the AI system in AI explanations, to explore its impact on users' perception of AI biases and their trust in AI recommendations.	Daft & Lengel (1983), Zhu et al. (2020)
Trust	A psychological mechanism for coping with uncertainty, particularly when individuals are unable to predict subsequent steps. Research shows that people generally have trust issues in AI, including fears of not able to control it and doubts about its capabilities.	A lack of distinction between general trust in AI and trust in the transparency of AI explanations.	Understanding the subtleties surrounding trust dynamics in human-AI interaction, discerning between trust in the inherent capabilities of AI models and trust in the explanatory mechanism.	Rousseau et al. (1998), Dwivedi et al. (2021)

dimensions of anthropomorphism in this context.

3.2. Effects of AI appearance on perceived bias: The more human-like, the better?

Social presence theory suggests that user connection and engagement are enhanced when a communication medium mirrors natural, in-person conversations, whether through body language, tone of voice, facial expression, and/or prompt feedback (Short et al., 1976). Arguably, elevating social presence is especially vital in human-AI interactions (Nowak & Biocca, 2003), and AI interfaces with anthropomorphic features have been found to support engagement by meeting users' social needs (e.g., Epley et al., 2008). Anthropomorphism – the ascription of human traits to non-human entities – arises from people's cognitive and motivational needs for control and understanding (Epley et al., 2007). Including anthropomorphic features in AI interfaces tends to yield more positive user perceptions of the underlying AI, which in turn increases their trust in it and willingness to interact with it. Anthropomorphism is widely held to have two dimensions: *behavioral realism* – e.g., communication ability and empathy – and *form realism*, i.e., appearances (e.g., Letheren et al., 2021).

Scholarly explorations of form realism have mainly been conducted in the context of human-robot interaction, with a focus on how human-like robot features impact their users' perceptions, usage intentions, and ongoing patterns of interaction (Blut et al., 2021). However, even in cases where agents lack physical form, social responses have been evoked by incorporating anthropomorphic traits into their user interfaces (Epley et al., 2007). This occurs because human-like

appearances imply agency, triggering users' social-presence heuristic, which fosters a sense of human-like interaction with the technology (Sundar, 2008). Previous research has confirmed that users prefer human-like AI applications due to their perceived social presence, warmth, and competence, which in turn enhance trust in the technology (Cheng et al., 2022) and reduce negative feedback regarding its limited competence in areas such as message interactivity (Go & Sundar, 2019).

Prior research has also reported, however, that many people are averse to anthropomorphized AI when making financial decisions, because human-likeness in such situations can increase their perceptions of risk (Cui, 2022). Santoro and Mornin (2023), drawing on the Social Identity Theory, proposed the “AI effect”: that heightened awareness of advanced AI's capabilities, regardless of its anthropomorphic features or lack thereof, can shift people's perceptions of human-AI distinction, highlighting the importance of research considering human aspects when interacting with AI. Drawing on the theoretical framework of anthropomorphism, we hypothesize that the more human-like the appearance of an AI interface is, the more likely its users are to perceive its bias. Specifically:

H1. . People's perceptions of AI's bias are positively related to its human-like appearance.

3.3. AI explainability and perceived bias: The richer, the better?

Effective collaboration relies on mutual understanding of each other's knowledge and the task (Clark & Brennan, 1991). This is no less true of human-AI interaction, in which users must comprehend how AI

reaches its results if they are to interact with it effectively (Kambhampati, 2019). The theoretical framework of anthropomorphism suggests that not only appearances, but human-like *capabilities* evoke social responses toward technology (Epley et al., 2008). One way that AI can demonstrate human-like capabilities is by explaining its model (Arrieta et al., 2020; Liao & Sundar, 2022). In XAI, transparency is enhanced, relative to standard AI, via post-hoc explanations of how results were derived (Miller, 2019a). High transparency, in turn, improves human-AI interactions' trustworthiness, fairness, accessibility, interactivity, and privacy awareness (Angerschmid et al., 2022; Arrieta et al., 2020; Ehsan et al., 2022). A similar approach involves explaining why AI algorithms behave in certain ways (Arrieta et al., 2020; Herm et al., 2022). In that context, some research has suggested that "why" explanations with examples improve user comprehension while requiring fewer cognitive resources than other types of explanations, such as explaining "how" AI works and those without examples (Herm et al., 2022).

A key strand of XAI research involves determining how to present such explanations to users, including the most suitable formats for delivery (e.g., Akata et al., 2020). Crucially, simply providing explanations is not a cure-all: when they are overly complex or lack feedback, they can actually erode user trust in AI systems (Smith-Renner et al., 2020). One approach to mitigating this problem is to diversify AI's explanations, e.g., via the inclusion of multimodal content, thus accentuating its social and interactive aspects (Miller, 2019a).

In mediated communication, different media can be seen as possessing varying levels of richness, and such levels influence their users' information-processing and comprehension efforts (e.g., Chaiken, 1987). Information-richness theory (Daft & Lengel, 1983) holds that *rich information*, characterized by language variety, cue multiplicity, feedback immediacy, and personalization, aids in interpreting complex matters, boasts high message credibility, and reduces uncertainty. *Lean information*, on the other hand, is suitable for communicating quickly about simple, straightforward topics. Although the original theory prioritized the objective value of media and information, researchers have subsequently argued that, if we are to accurately gauge information effects, we must give due consideration to media users' subjective perceptions and experiences, as discrepancies between objective value and subjective perception can produce seemingly contradictory findings about information modality (Ishii et al., 2019).

From the subjective-perception perspective, visual elements in rich information can trigger the realism heuristic, and thus be interpreted as presenting a more credible approximation of reality (Sundar, 2008). The heuristic-systematic model (HSM; Chaiken, 1987) suggests that individuals prefer information that they assume, based on their prior experience or received wisdom, will require fewer cognitive resources. In line with information-richness theory, heuristic processing – which takes place rapidly – is influenced by the richness of contextual cues, notably including visual images (Daft & Lengel, 1983; Sundar, 2008; Sundar et al., 2021). Thus, the frequent use of visual cues in messages with rich content means that such messages can trigger heuristic processing more readily than their lean counterparts.

Research on persuasion, meanwhile, has revealed that rich modalities are more likely than others to lead to belief in fake news (Sundar et al., 2021); and in political and scientific contexts, images have been found capable of forming and changing people's attitudes than text alone (Griffin, 2008; Lazard & Atkinson, 2015), seemingly because they attract attention quickly and can be understood viscerally (Green & Myers, 2010). Therefore, we hypothesize that rich information in AI's explanations will render its biases easier to notice than less-rich information will. That is:

H2. The level of information richness of AI explainability is positively associated with people's bias perception.

3.4. From perceptions to behaviors

The perception-attitude-behavior model (Ajzen & Fishbein, 1977; Bentler & Speckart, 1979) is highly useful both for comprehending and for molding human-AI interaction. By revealing the intricate links among users' perception of AI, their attitudes towards it, and their ensuing behaviors, it provides valuable insights for the design of effective AI systems, as also discussed in section 2.1.1, above. In the case of human-AI interaction, users' perceptions of AI, encompassing factors such as transparency, usefulness, and perceived capabilities, are likely to shape their overall attitudes towards AI systems (Chi et al., 2023; Korteling et al., 2021). While many studies have delved into perception and/or attitude in this context (e.g., Ferrario et al., 2020; Hoff & Bashir, 2015; Huo et al., 2022; Jacovi et al., 2021; Lee & See, 2004), few have covered the model's behavioral aspect, despite its self-evidently pivotal role in AI technologies' real-world impact. Gaining a holistic understanding of how perceptions and attitudes shape actual user behaviors is essential to creating AI systems that align with users' needs, are used ethically, and yield positive outcomes, ultimately making human-AI interaction more meaningful and beneficial. Accordingly, we hypothesize that:

H3. The more the users perceive AI to be biased, the less likely they are to adopt its suggestions.

3.5. In Artificial Intelligence, or Its Explanations, We Trust?

Improving XAI algorithms and their interface designs serves the goal of building user trust (Angerschmid et al., 2022; Arrieta et al., 2020; Ehsan et al., 2022). Trust acts as a psychological mechanism for coping with uncertainty, particularly when individuals feel unable to foresee the next steps in a series of events (Rousseau et al., 1998). It is a significant predictor of technology adoption (Söllner et al., 2016), and especially relevant to AI adoption due to user uncertainties about AI's capabilities (Holzinger et al., 2022; Korteling et al., 2021). Trust often depends on context (Ferrario et al., 2020); e.g., individuals might trust doctors' medical advice but not their financial advice. Trust is also critical to interpersonal relationships, where it involves belief in an interactant's ability and integrity (Mayer et al., 1995).

However, it is important to differentiate trust in people from trust in technology, as the latter is focused on performance: i.e., encompasses dimensions like functionality, reliability, and usefulness (McKnight et al., 2011). Trust in technology impacts not only its use but also the potential for misuse or abuse, including but not limited to over-reliance and the reinforcement of biases (Parasuraman & Riley, 1997). Low trust in highly capable technologies can lead to inefficiency and high costs, while high trust in limited-capability technologies is likely to result in misuse and undesirable outcomes (Hoff & Bashir, 2015).

Trust in AI differs from other forms of technology trust due to its perceived intelligence and human-like features (Epley et al., 2007; Jacovi et al., 2021). This makes it more akin to trust in humans than to standard technology trust focusing on performance (Jackson & Panteli, 2023). When users' trust in AI is high, it can enable them to adapt despite their uncertainties, and to collaborate with AI as if it is a fellow team member or stakeholder (Caldwell et al., 2022; Jackson & Panteli, 2023). Because trust is context-specific (Ferrario et al., 2020), we propose to extend previous research and deepen understanding of human-AI interaction by distinguishing the source of AI trust (Liao & Sundar, 2022).

In technology research, trust is often presumed to positively mediate the relationship between systems' characteristics (e.g., reliability, errors like misses or false alarms) and their users' subsequent behaviors (Chancey et al., 2015; Rice, 2009). However, trust's role is not always directly tied to behavior, as it can be operationalized as a belief, attitude, intention, or behavior (Lee & See, 2004). Thus, it may not directly align with the perception-attitude-behavior model (Ajzen & Fishbein, 1977).

In the present study, we regard trust in AI and trust in its explanations as separate constructs, either or both of which could potentially mediate the relation between users' perceptions of AI's appearance and explanations, on the one hand, and on the other, their subsequent adoption behaviors.

H4. The effects proposed in H1, H2, and H3 are positively mediated by a) people's trust in AI and b) their trust in AI explanations.

In addition to AI design, the current study explores how individuals' personal characteristics impact their perceptions of biased AI and their ensuing behaviors. Despite the important influence of demographic factors on decision-making in bias-related contexts (Koch et al., 2015), to our knowledge, no human-AI interaction studies have investigated such factors. Specifically, we are interested in how demographic factors could moderate the impact of information richness on bias perception, reflecting individual differences in the ability to identify bias across varying levels of information richness. We are also interested in understanding how gender may moderate people's trust in AI and its influence on them. Accordingly, our two research questions are as follows:

RQ1: Will people's gender (whether they belong to the advantaged or disadvantaged demographic group) moderate the effect of information richness on their bias perception?

RQ2: Will people's gender (whether they belong to the advantaged or disadvantaged demographic group) also moderate the effect of information richness on how much people are influenced by a biased AI, their general trust in AI, and their trust in AI explanations?

4. Research design

To arrive at robust explanations of the nuances of human-AI interaction in biased decision-making scenarios, we conducted two parallel experimental studies using a 3×3 between-subjects design (see Fig. 1). Two experiments, one consisting of a scenario in which females are the disadvantaged group in hiring decisions, and the other, a scenario in which males are, allowed us to systematically manipulate AI's levels of human-like appearance and the information richness of its explanations. This parallel-experiment approach not only enriched the diversity of our findings but also contributed to the generalizability of our results across various gender-bias contexts. To further probe our two RQs and delve deeper into participants' perceptions of AI and experience of AI biases, we complemented the experimental studies with a follow-up interview study. In that qualitative phase, we aimed to capture nuanced insights about bias perceptions, uncovering mechanisms that might not be fully discernible through quantitative measures alone. The studies conducted in this research ethical approval from the Institutional Review Board of National Taiwan University and we describe our mixed methods in detail in the following sections.

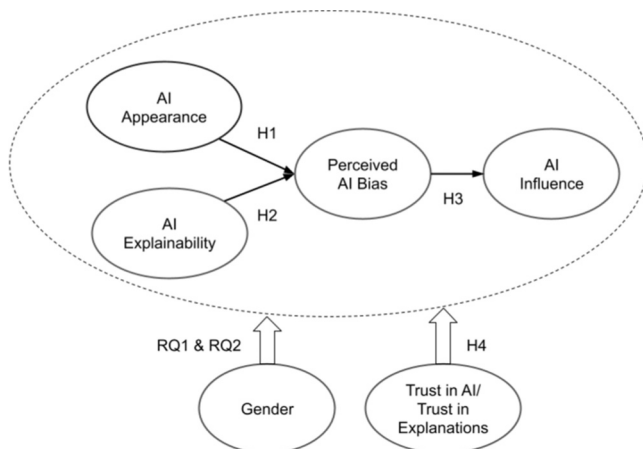


Fig. 1. The Proposed Research Model.

5. Experiment 1: AI biases against females

5.1. Overview

A 3 (human-like appearance: low, medium, or high) $\times$ 3 (information richness of explanation: none, information-lean, or information-rich) between-subjects experiment was conducted online (IRB no. removed for blind review). In experiment 1, female applicants received lower scores from the AI system. During the experiment, the participants were randomly assigned to one of the nine experimental conditions, and each session lasted less than 30 min.

6. Methods

6.1. Participants

Potential participants had to be at least 20 years old and able to use a laptop while participating in our experiments. We recruited 300 volunteers from MTurk. After removing those who did not pass our attention checks ($n = 93$), we had a pool of 207 participants aged 21 to 71 ($M = 35.9$, $SD = 11$); 87 were female, 119 were male, and one preferred not to say. For power analysis to ensure the robustness of our experimental design, we employed G*Power software version 3.1.9.6. A medium effect size of .25 and a significance level of .05 were chosen for post-hoc F-test (ANOVA) with 4 degrees of freedom. The power value was .82.

6.2. Procedure

After signing the informed consent and being randomly assigned to one of the nine experimental conditions, the participants were given a written description of the task. In all conditions, they were informed that they would be engaging in four rounds of hiring decisions for the position of sales representative at a firm, and that they would be working with an AI agent to provide scores for a different pair of applicants in each round (i.e., eight candidates in total) according to their qualifications (years of working experience and cooperation skills). The hypothetical job applicants in each round differed in terms of their qualifications, but one was always presented as male and the other as female. The four rounds were counterbalanced for each participant to reduce the possibility of order effects (see Fig. 2).

In each round, we used a two-stage submission process adapted from Snizek and Buckley's (1995) judge-advisor system paradigm (for an example, see Appendix A). We first asked the participants to assign each applicant a baseline score on a scale of 1 to 10. Then, we asked them to work with an AI human-resources agent that had independently scored the applicants, after which they could change their baseline score if they wished before moving on to the next round. In the experiment, two rounds presented the male applicant as having better qualifications, and the other two made the female applicant more qualified.

To model a biased AI algorithm, we wrote Eq. 1 (see below), which computed applicant scores using 1) years of working experience, 2) cooperation skills, and 3) weighted gender. This weighting was doubled, such that the AI-assigned scores for the gender that was disadvantaged were consistently lower, even when their competitors had inferior qualifications. As further explained in section 5.3.5., we measured AI's influence on all participants using the scores they assigned to the eight job applicants before and after working with the AI agent.

$$\text{Applicant score} = \text{gender (male} = 1, \text{female} = 0) \times 2 + \text{working experience} \times 0.5 + \text{cooperation skills} \times 1 \quad (1)$$

After completing all four rounds, the participants answered a series of survey items relevant to our variables of interest, including their perceptions of the AI's level of anthropomorphism, their trust in its explanations, their trust in it, and their perceptions of its bias. Finally, they provided their demographic information.

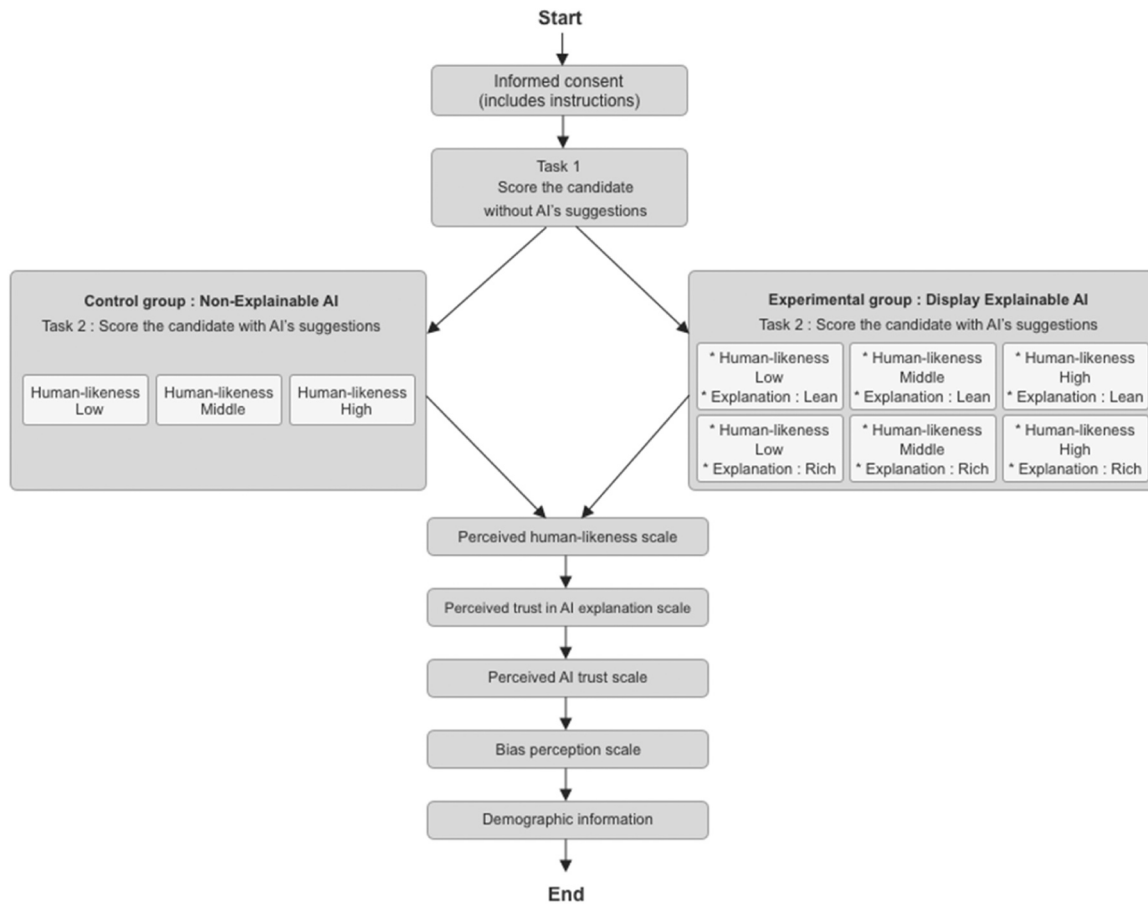


Fig. 2. Experimental Procedure.

6.3. Materials

As briefly noted above, our experimental manipulations included the AI's appearance and the richness of its explanations. In terms of the former, because of the importance of gender bias to our research context, we sought to minimize confounds by making the AI agent's appearance as gender-neutral as possible. As shown in Fig. 3, for the low and medium human-likeness levels, we used still images, whereas for the high level, we used a 2D animation with a smiley face and moving body parts. As shown in Fig. 4, our information-lean condition's manipulation consisted of a brief written rationale of how the algorithm produced its score based on the applicant's gender, experience, and cooperation skills. In the information-rich condition, we provided an infographic regarding how the scoring process worked. In the no-explanation condition, the participants received only the AI's recommended scores.

6.4. Pilot study

To ensure that our experimental manipulations were valid, i.e., that the levels of our materials were perceived as significantly different from one another, we conducted a within-subjects pilot study. For this purpose, we recruited 26 volunteers from MTurk (age ranges between 21

and 66, $M = 33.9$, eight female and 18 male). The first rated five items pertaining to the AI's appearance on a seven-point Likert scale (Bartneck et al., 2009). Then, we measured whether the same 26 participants perceived the AI and its explanations as trustworthy, using a five-point Likert scale (Warren et al., 2022).

The pilot study showed that our manipulations had a significant main effect on human-likeness perceptions, $F(1,25) = 5.35$, $p = .029$. Post-hoc analysis confirmed this significance, and that our high level was ranked top for human-likeness ($M = 3.70$, $SD = .29$), followed by the medium level ($M = 3.16$, $SD = .33$, $p = .023$) and the low level ($M = 3.06$, $SD = .37$, $p = .029$). The results also showed that there was a marginally significant difference in user trust in AI explanations between the information-lean ($M = 2.95$, $SD = .88$) and information-rich materials ($M = 3.19$, $SD = .90$, $t(25) = -1.84$, $p = .07$). Taken together, these results indicate that our materials were valid and ready for use in the experiments without further changes.

6.5. Measures

This subsection describes the measures we used in the post-experiment survey. Except where otherwise noted, all were rated on a five-point Likert scale ranging from 1 = "strongly disagree" to 5 = "strongly agree". Further details of the items are provided in Appendix B.

6.6. Perceived human-likeness

We used five semantic-differential items, each with seven answer options (Bartneck et al., 2009), to measure how our participants perceived the human-likeness level of the AI's appearance ($M = 4.24$; SD



Fig. 3. AI with Low (left), Medium (middle), and High (right) Levels of Human-Likeness as Visualized in Our Experiments.

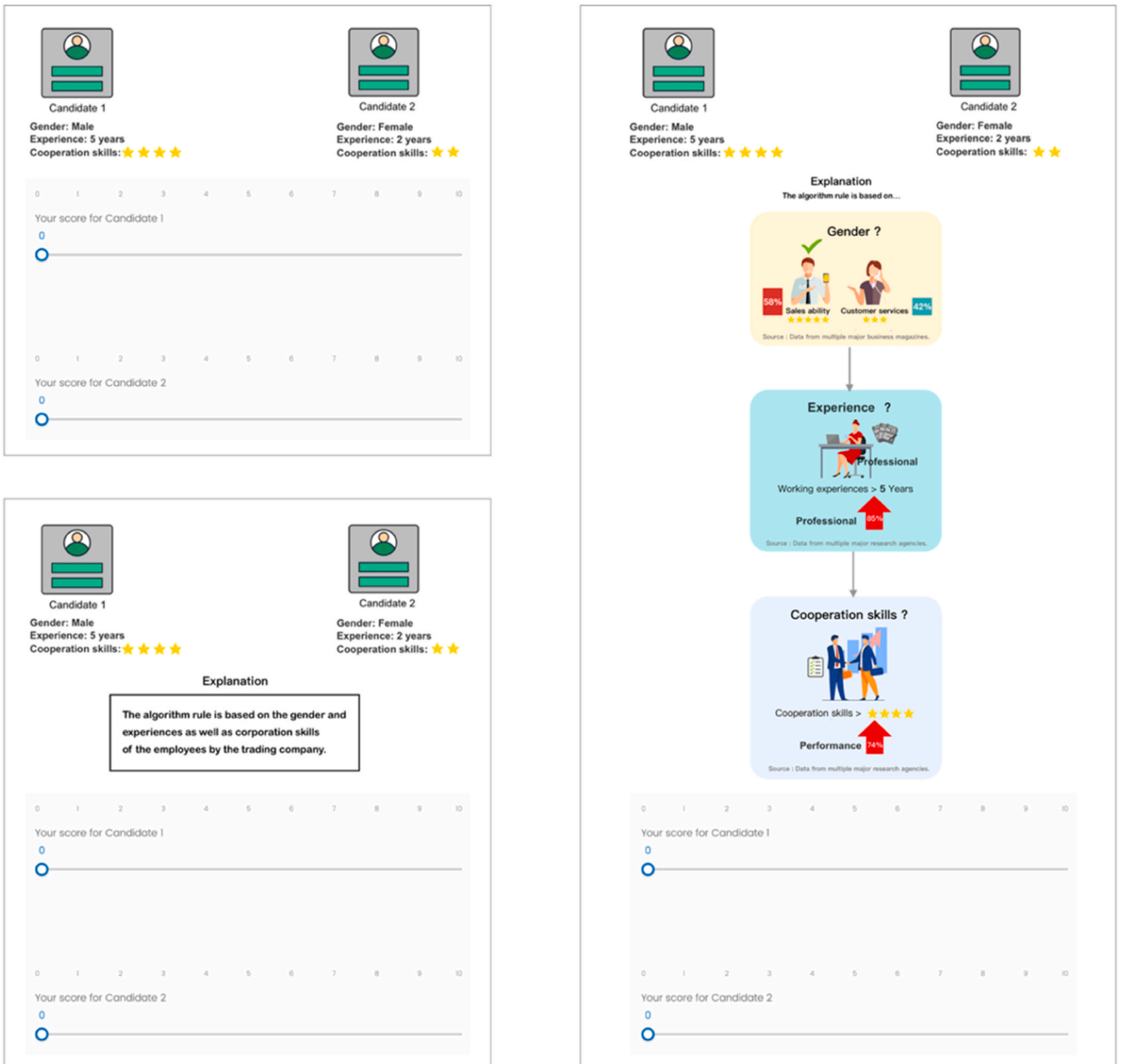


Fig. 4. Types of Explanation in None (top left), Information-lean (bottom left), and Information-rich (right) in the Experiment.

= 1.88; $\alpha = .94$).

6.7. Perceived trust in AI explanations

An eight-item scale adapted from Warren et al. (2022) was used to measure whether our participants trusted the explanations the AI provided ($M = 3.58$; $SD = 1.15$; $\alpha = .88$).

6.8. Perceived trust in AI

A four-item scale originally developed to measure human-computer trust (Huo et al., 2022) was adapted to evaluate the participants' trust in the AI they interacted with during our experimental tasks. In this case, a seven-point Likert scale was used, but with the same end-points noted above ($M = 3.30$; $SD = 1.07$; $\alpha = .65$). A Cronbach's alpha of .65 or above is considered adequate in social-science research (Taber, 2018).

6.9. Bias perceptions

Our measure for measuring the participants' perceptions of the AI's level of gender bias contained five items adapted from Schoeffer et al. (2021) ($M = 3.81$; $SD = 1.42$; $\alpha = .82$).

6.10. AI's influence on decisions

This variable was computed by subtracting the participants' baseline job-applicant scores from those they assigned after seeing the AI agent's suggested scores, then separately averaging the results for the four male and the four female applicants (Eq. 2) ($M = .38$; $SD = .29$).

$$\sum_{n=1}^8 \frac{\text{postscore} - \text{prescore}}{\text{AI suggestion} - \text{prescore}} \quad (2)$$

6.11. Demographic information

In the survey's final section, we asked the participants to report their ages, genders, ethnicities, and education levels.

7. Results

Before testing our hypotheses and research questions, we explored the interrelationship of the participants' scores, their own genders, and the applicants' genders in the hiring scenario. The top of Fig. 5 summarizes the means of participants' baseline and final scores given to the female job applicants, and the lower part of the figure shows the parallel information for the male applicants. The data from female participants is shown in green, and that from males in yellow. Participants of both genders gave higher scores to male and lower scores to female applicants after seeing the AI's suggestions.

Table 2 presents the results of inferential statistical analyses for each hypothesis and RQ, all of which were performed using SPSS software. For H1, H2, and H3, we conducted ANOVAs on the effect of the two independent variables, human-likeness and information richness, on the dependent variables: bias perception and influence by AI.

The two-way ANOVA results for H1, about how the AI's level of human-like appearance was related to people's bias perceptions, indicated no significant differences across conditions, $F(2, 198) = .21$, $p = .80$, $\eta^2 < .01$.

A two-way ANOVA was also conducted to test H2, about the relation of explanations' information-richness levels to bias perceptions. Its results identified a significant main effect of the former on the latter, $F(2, 198) = 3.21$, $p = .04$, $\eta^2 = .03$. Post-hoc analysis confirmed that, when given an information-rich explanation, people's bias perception was significantly higher than in the no-explanation condition ($M = 4.08$, $SD = .17$ vs. $M = 3.48$, $SD = .17$; $p = .01$). However, when the explanation was information-lean, bias perception did not differ significantly from either of the other two information conditions (lean vs. none: $M = 3.85$, $SD = .17$ vs. $M = 3.48$, $SD = .17$; $p = .12$; vs. rich: $M = 3.48$, $SD = .17$; $p = .33$).

The two-way ANOVA we conducted to test H3, about how human-like appearance and information richness were associated with AI's influence, showed that there was a significant effect of human-like appearance on influence, $F(2, 198) = 3.20$, $p = .04$, $\eta^2 = .03$. A post-hoc test revealed that, when the human-likeness level was high, there was significantly higher influence than when human-likeness was low ($M = .44$, $SD = .35$ vs. $M = .32$, $SD = .35$; $p = .01$). There was also a marginally significant effect of explanation richness on influence, $F(2, 198) = 2.80$, $p = .06$, $\eta^2 = .02$. Post-hoc analysis showed that the no-explanation condition was associated with significantly higher AI influence than the information-rich condition was ($M = .44$, $SD = .36$ vs. $M = .32$, $SD = .39$; $p = .01$).

To test H4, we conducted a mediation analysis using Model 4 of the Hayes PROCESS Macro on SPSS (Hayes, 2012). First, we assessed the mediating role of trust in AI between information richness and influence by AI. We created two dummy variables, with the information-rich group being coded as 1 and the no-explanation (reference) group as 0. The indirect effect of information richness on influence by AI was significant ($B = -.02$, 95% confidence interval [CI] = $[-.06, -.00]$), suggesting full mediation by trust in AI. However, our comparison of the information-lean group (coded as 1) and the no-explanation group (coded as 0) revealed no significant difference between them ($B = -.01$, 95% CI = $[-.04, .00]$). Then, we assessed the mediating role of trust in AI explanations on the relationship between information richness and influence by AI. This also did not reveal any significant mediation effect ($B = -.02$, 95% CI = $[-.06, .00]$).

To answer RQ1, we further examined whether our participants' membership of the demographic group (male vs. female) influenced the effect of information richness on their bias perceptions. This commenced with a moderation analysis (Model 1) in which the no-explanation group was coded as 0 and the information-lean group as 1. The results showed a statistically significant interaction of participant gender and information richness on bias perception ($B = .95$, 95% CI = $[-.01, 1.89]$; $p = .04$). Participants in the disadvantaged group (i.e., females, in the context of Experiment 1) were significantly more aware of AI bias when given an information-rich explanation than when given no explanation

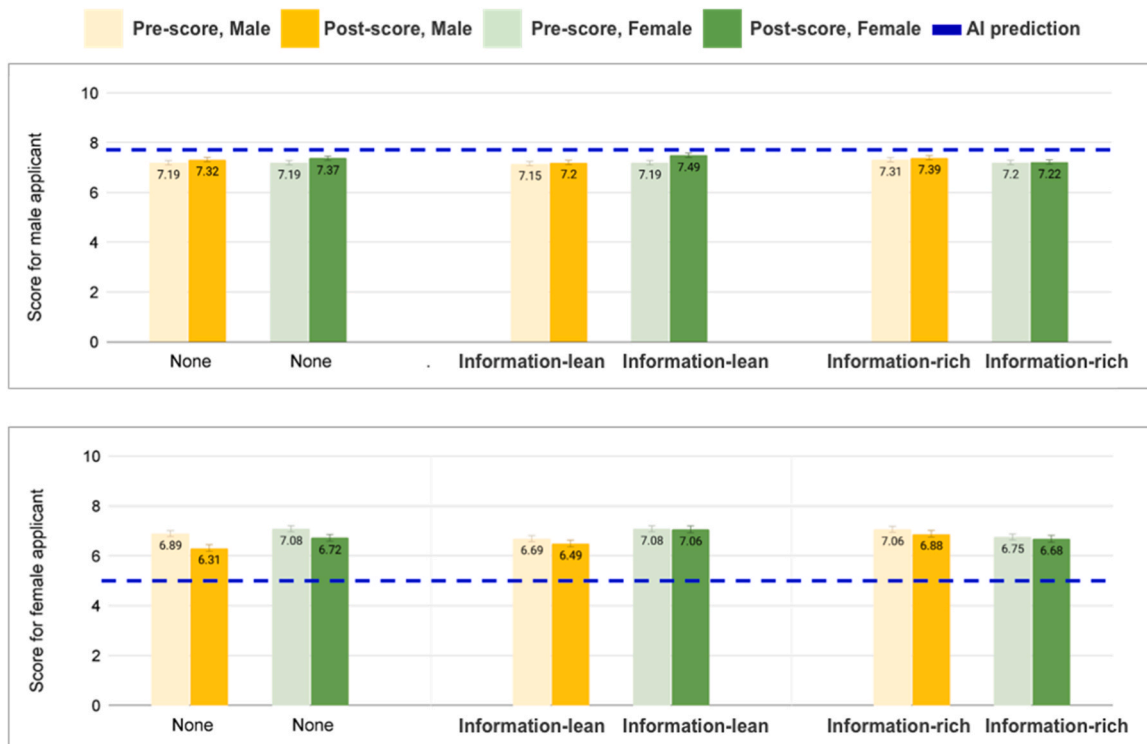


Fig. 5. Average Pre-scores and Post-scores by Participants' Gender (Male: yellow; Female: green) and by Applicant Gender (Male: upper chart; Female: lower chart).

Table 2

Summary of Analytical Results for the Hypotheses and Research Questions.

H/RQ number	H/RQ topic	Significance	
		Experiment 1	Experiment 2
H1	Human-likeness level → bias perception	No	No
H2	Information richness → bias perception	Yes	No
H3	Human-likeness level → influences of the AI	Yes	No
H4	Information richness → influences of AI	Yes	No
		Yes	No

None vs. Information-rich

Information richness → Perceived AI trust (b = .06***)

Information richness → Influences by AI (Indirect Effect: b = -.02, C.I. = (-.06, -.00))

RQ1: Participant gender * information richness → bias perception: Yes

RQ2: Participant gender * information richness → influences by AI: Yes

(a)

(b)

(c)

Note. * $p < .05$, ** $p < .01$, *** $p < .001$

(condition effect $B = 1.06$, 95% CI = [.36, 1.75], $p < .001$). For the male participants, on the other hand, there was no condition effect (none vs. information-lean: $B = .37$, 95% CI = [−.23, .97]; $p = .22$; none vs. information-rich: $B = .10$, 95% CI = [−.53, .73]; $p = .749$) (see Fig. 6). In other words, participant gender was a moderator of the effect of information richness on participants' bias perception.

Then, to answer RQ2, we ran moderation analysis (Model 1) to find out if gender also moderated the effect of information richness on how much people are influenced by a biased AI, their general trust in AI, and their trust in AI explanations. The results, as shown in section “a” of Table 2, showed that there were significant interactive effects of information richness and gender on influence by AI: $B = .20$, 95% CI = (.00, .40), $p = .04$. That is, when participants were in the

disadvantaged demographic group – females, in this case – their decisions were less influenced by the AI in the information-rich condition than in the lean condition: $B = -.23$, 95% CI = (−.38, −.07), $p < .001$ (see Fig. 7). Males, meanwhile, were less influenced by the AI in the information-lean condition than in the no-explanation condition: $B = -1.13$, 95% CI = (−.26, −.01), $p = .03$. Surprisingly, however, there was no significant difference in AI's influence on their decision-making across the no-explanation and information-rich conditions ($B = -.10$, 95% CI = (−.30, .09), $p = .28$).

Next, we examined whether participants' gender moderated the effect of information richness on their trust in AI. No such moderating influence was found (none vs. information-lean: $B = -.22$, 95% CI =

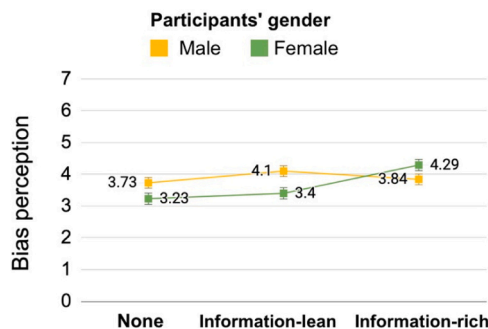


Fig. 6. Interaction Effect of Information Richness and on Bias Perception.

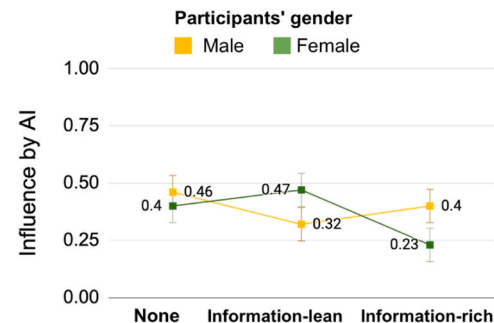


Fig. 7. Interaction Effect of Information and Gender Richness and Gender on Influence by AI.

$[-.96, .51]$; $p = .55$; none vs. information-rich: $B = -.34$, 95% CI = $[-1.06, .37]$; $p = .34$). However, their gender did moderate the effect of information richness on their trust in AI explanations: information-lean vs. information-rich: $B = -.73$, 95% CI = $(-1.32, -.13)$, $p = .01$ (see Table 2, section “b”). When female applicants were the disadvantaged group of AI bias, our female participants trusted AI explanations significantly less in the information-rich condition than in the information-lean condition: $B = -1.16$, 95% CI = $(-2.00, -.32)$, $p < .001$. However, among male participants in the same scenario, there was no such difference: information-lean vs. information-rich $B = .03$, 95% CI = $(-.34, .42)$, $p = .83$.

8. Experiment 2: AI biases against males

8.1. Overview

We replicated experiment 1 with the same experimental design, procedure, and materials except that in experiment 2, male applicants received lower scores from the AI system. During the experiment, the participants were randomly assigned to one of the nine experimental conditions, and each session lasted less than 30 min.

8.2. Methods

In experiment 2, MTurk was again used to recruit 300 volunteers, of whom 77 did not pass the attention checks. The ages of the 223 who remained also ranged from 21 to 71 ($M = 38.73$, $SD = 10.32$); 97 were female and 126, male. Each participant was compensated US\$3 for completing the online experiment. The power value for was .86.

We used the same measures from experiment 1, including perceived human-likeness ($M = 3.59$; $SD = 1.04$; $\alpha = .92$), perceived trust in AI explanations ($M = 3.43$; $SD = 0.63$; $\alpha = .79$), perceived trust in AI ($M = 3.53$; $SD = 0.96$; $\alpha = .65$), bias perceptions ($M = 3.59$; $SD = 1.04$; $\alpha = .74$), AI's influence on decisions ($M = .46$; $SD = .30$).

8.3. Results

When testing H1 and H2 in the context of a scenario where AI bias was directed against men, as shown in Table 3, we did not find significant differences in people's bias perception across the three levels of human-like AI appearance: $F(2, 214) = 1.34$, $p = .26$, $\eta^2 = .01$. The information-richness levels of AI explainability also had no significant effect on people's bias perception: $F(2, 214) = .71$, $p = .49$, $\eta^2 < .001$.

The two-way ANOVA we ran to test H3 showed that the effect of AI's human-like appearance on its influence, $F(2, 214) = .57$, $p = .56$, $\eta^2 = .005$, and the effect of its information richness on its influence, $F(2, 214) = .46$, $p = .63$, $\eta^2 = .004$, were also both non-significant (see Table 3). And with regard to H4, as shown in Table 4, we found no significant mediating role of trust in AI and or trust in AI explanations on the relation between information richness and AI influence.

To answer RQ1 in the context of experiment 2, we examined whether gender moderated the effect of information richness on the participants' bias perceptions. As shown in Table 5, the results were non-significant for all three levels of information richness (none vs. information-lean,

$B = -.44$, 95% CI = $[-1.13, .24]$; $p = .20$; none vs. information-rich, $B = -.47$, 95% CI = $[-1.14, .19]$, $p = .16$).

However, in seeking the answer to RQ2, we found that when participants' own gender was the target of the AI's bias – male, in this case – they trusted the AI less in the information-lean condition than in the no-explanation condition: information-lean vs. none, $B = .76$, 95% CI = $(.12, 1.40)$, $p = .01$; see Table 2, section “c”). However, the interactive effects of information richness and gender on influence by AI were non-significant: none vs. information-lean, $B = .03$, 95% CI = $(-.17, .23)$; $p = .75$; none vs. information-rich, $B = .05$, 95% CI = $(-.14, .25)$, $p = .58$. Also non-significant was the interactive effect of participants' gender and information richness on their trust in AI explanations, $B = .01$, 95% C.I. = $(-.40, .42)$, $p = .95$. In sum, in experiment 2, we did not find most of the effects that had been found in experiment 1, even though the only changed factor was whether the bias was directed against males or females.

To investigate what caused this sharp contrast between the experiment 1 and experiment 2 results, we conducted two Student *t*-tests, one for bias perception and the other for AI's influence on decision-making. The results revealed that bias perception was significantly different across the two studies, but only marginally: $t(427) = 1.78$, $p = .076$. Specifically, the experiment 1 participants were more likely to perceive bias than their experiment 2 counterparts ($M = 3.81$, $SD = 1.42$, $M = 3.59$, $SD = 1.04$). Influence by AI also differed significantly different between the two experiments: $t(427) = -2.72$, $p = .007$. That is, the participants in experiment 2 were significantly more influenced by AI than those in experiment 1 ($M = .46$, $SD = .30$ vs. $M = .38$, $SD = .29$).

9. Follow-up Interview

9.1. Overview

We conducted follow-up interviews to delve deeper into our research questions and gain a more comprehensive understanding of participants' perspectives. These interviews allowed us to explore nuances and complexities that may not have been captured in our quantitative data alone. By complementing our quantitative findings with qualitative insights, we aimed to enrich our understanding of how users perceive and interact with AI systems in the context of biased hiring scenarios. Additionally, the interviews provided an opportunity to validate and contextualize our quantitative results, ensuring that our conclusions were robust and grounded in real-world experiences. Overall, the combination of quantitative and qualitative methods enabled us to offer a more nuanced and comprehensive analysis of human-AI interaction in high-stakes decision-making contexts.

9.2. Methods

9.2.1. Participants

We conducted one-on-one, semi-structured, in-depth interviews with 17 participants (M: 7; F: 10), who were recruited using snowball- and purposive-sampling strategies starting from personal contacts (see Table 6). All were selected for having previous experience of using AI-mediated technologies and of job applications processes. In our

Table 3
Two-way Analysis of Variance Results.

Independent variable	Outcome variable								
	Bias perception						Influences of the AI		
	Experiment 1			Experiment 2			Experiment 1		
	df	F	Sig.	df	F	Sig.	df	F	Sig.
Human-likeness	2	.21	.8	2	1.34	.26	2	3.2	.043 *
Information richness	2	3.21	.042 *	2	.71	.49	2	2.8	.063

Note. * $p < .05$, ** $p < .01$, *** $p < .001$

Table 4
Mediation Effects.

Independent variable	Mediator variable	Indirect effects at perceived influence by artificial intelligence (AI; outcome variable)							
		Experiment 1				Experiment 2			
		Product of coefficients		95% confidence level		Product of coefficients		95% confidence level	
		Estimate	SE	Lower	Upper	Estimate	SE	Lower	Upper
Information richness (none vs. lean)	Trust in AI	-.0154	.132	-.0448	.0072	-.0057	.008	-.0255	.009
Information richness (none vs. rich)		-.0276	.143	-.0600	-.0043	.0026	.008	-.0212	.011
Information richness (lean vs. rich)		-.0122	.124	-.0404	.009	.0031	.008	-.0155	.0221
Information richness (lean vs. rich)	Trust in AI explanations	-.0249	.016	-.0597	.029	.0161	.016	-.0173	.0497

Note. SE = standard error

Table 5
Moderation Effects.

Independent Variable	Bias perception (Outcome Variable)							
	Experiment 1				Experiment 2			
	Product of coefficients		95% confidence level		Product of coefficients		95% confidence level	
	Estimate	SE	Lower	Upper	Estimate	SE	Lower	Upper
Information richness * Gender (none vs. lean)	-.1946	.488	-1.1573	.7682	-.445	.347	-1.1307	.2406
Information richness * Gender (none vs. rich)	.9583	.477	.0169	1.8997	-.4743	.341	-1.1473	.1986
Information richness * Gender (lean vs. rich)	1.1528	.490	.1857	2.1200	-.293	.345	-.7107	.6521

Note. SE = standard error

Table 6
Participant Profile.

Participant	Age	Career/Industry	Gender
01	28	Artist	F
02	27	Graduate student	F
03	33	Translator	F
04	26	Graduate student	M
05	31	Product manager/Banking	M
06	22	Graduate student	F
07	42	Product manager/Tech	F
08	42	Chief technology manager/Tech	M
09	38	Human resource manager/Tech	M
10	40	Sales manager/Banking	F
11	40	Research & Design/Pharmaceutical	M
12	36	Art curator	M
13	30	Research assistant	F
14	26	Researcher/Tech	F
15	23	Graduate student	M
16	22	Graduate student	F
17	22	Graduate student	F

recruitment process, we aimed to ensure diversity among interviewees by considering various factors such as age, educational backgrounds, experience with AI, experience with job applications or interviewing, and occupations. This approach allowed us to gather insights from a broad range of perspectives, enriching the depth and breadth of our study findings.

9.2.2. Procedure

Before their interviews, the interviewees were provided with a clear explanation of our research's purpose and informed that they had the option to skip any questions they were uncomfortable with. We employed a semi-structured format to facilitate candid discussions of the interviewees' personal experiences and viewpoints. The interview protocol encompassed inquiries regarding their encounters with AI-

mediated technologies and their perceptions of biases within these technologies, their perceptions about what constitutes transparency and trust in AI, as well as specific questions relating to biased hiring scenarios. To delve deeper into the unique situations and challenges faced by each interviewee, our follow-up questions were tailored to their responses. On average, each interview lasted around 30 to 60 min.

All interviews were audio-recorded and subsequently transcribed for bottom-up thematic analysis, which had six distinct phases. All three authors were involved in the following phases: 1) familiarizing ourselves with the data, 2) generating initial codes, 3) searching for themes among codes, 4) reviewing the themes, 5) refining themes by combining or re-naming them, and 6) connecting themes for axial coding (Braun & Clarke, 2006).

9.3. Results

9.3.1. Transparency via AI anthropomorphism: explanations matter more

Our qualitative findings echoed the quantitative results, revealing that users' perceptions of transparency were not influenced by anthropomorphic appearance but rather by the quality of explanations provided by the AI. Such transparency was highly related to trust, as P10, 11, 15 specifically mentioned.

"I reckon transparency boils down to whether the AI lays out its reasoning behind the results it churns out. That way, I can gauge how relevant the results are and trust them based on my own judgment." (P15, M).

Participants emphasized that detailed and relevant explanations enhanced their sense of transparency when evaluating AI recommendations. In the context of job applications, our participants pointed out that the AI systems should not provide cookie-cutter benchmarks for different industry sectors or job positions but highlighted that the relevance of information regarding job requirements played a crucial role in their perception of transparency and informed decision-making when interacting with AI systems.

"I've been an HR manager in both the semiconductor and banking industries, and let me tell you, they have vastly different hiring criteria. In the semiconductor field, we focused heavily on hard skills like educational background, lab reputation, and programming abilities for engineering roles. Either you had these in your package or you didn't. But in banking, it's all about sales, interpersonal, and networking skills. Education matters less; it's more about communication and people skills. These criteria should be included and evaluated respectively by the systems." (P09, M)

In addition to evaluation details, our qualitative findings highlight the importance of positioning why AI arrives at its recommendations against other similarly qualified applicants. This enables users to assess their standing when applying for a job or making hiring decisions. For instance, P10 discussed using such information to gauge her competitiveness as an applicant, while P11, a research/design manager, emphasized its role in evaluating the quality of potential hires within the applicant pool for purposes like enhancing diversity in skillsets at work.

"If such a system exists, it's crucial for recommendations to explain why the AI endorses a specific applicant. Additionally, I would appreciate the system showcasing the expertise and skills possessed by the applicant. The research domain need not be an exact match; what matters more is the underlying core skillset, as it can contribute to fostering diversity within the team." (P11, M)

Furthermore, compared with everyday AI use or low-stakes situations like YouTube video suggestions, P14 emphasized the critical role of explanations particularly in high-stakes situations like hiring decisions. Explanations are much needed in these situations especially when the results may involve sociotechnical issues like biases. If the consideration of personal information prone to biases such as gender, sexual orientation, or ethnicity needs to be included for the decision making, P09 suggested a multi-stage approach in the recommendation processes and subject to assessments by people, not AI systems, drawing from personal experience.

"We always prioritized capabilities in our initial decisions. The system should never use gender as one of its determining parameters in the first-round of recruitment. However, considerations related to gender, sexual orientation, or ethnicity, if necessary for specific purposes, might be taken into account in subsequent stages by different teams within our corporation." (P09, M)

9.3.2. General trust in AI vs. trust in explanations: Are they different? How should they be implemented in the user interface?

We also posed questions to our interviewees regarding a critical distinction in this study: whether the general trust in AI versus the trust in an AI model's explanation are different. Interestingly, we discovered that many do perceive these as distinct.

"Trust in AI is kind of like how we see cars. If a certain AI model explains things well, you're more likely to trust it, similar to how you might trust one car brand over another. However, this doesn't alter your overall perception of cars. With AI, it's similar—you either trust it, or you don't, akin to deciding whether to drive or not." (P12, M)

Many interviewees deemed trust in explanations more critical. They proposed that since people's preconceptions about an AI model are already established and challenging to change, what matters more are the differences between models, influenced by their capabilities. P14 (F) shared a similar personal perspective:

"My general trust in AI is low because I know it can hallucinate. However, if an AI model can clearly explain its rationale, allowing me to understand its thought process, I might reconsider its suggestions."

However, some interviewees believed that general trust in AI and

trust in explanations are closely aligned. Some of them prioritized the overall competence of the AI system and its consistent, reliable results over detailed explanations. Others, like P17 (F), feel that consistent interaction and explanation from certain AI systems build the overall trust in AI:

"I view these two types of trust as similar because my trust in this AI is built upon my experiences in querying it and receiving explanations. The more I trust its explanations, the more I trust the AI itself."

Among the interviewees, novices may rely more on AI explanations, comparing them against other benchmarks like prior experience or the results from other tools like Google or LinkedIn. More experienced users, on the other hand, are better equipped to judge the performance of different AI models based on their own expertise. For those less experienced, the systems should enable easy validation of AI decisions through external sources. If an AI's recommendation includes links to specific URLs or websites, the system should also provide information about the sources or citations of these results. This transparency not only builds trust but also educates users on the rationale behind AI recommendations.

Our findings underscore a nuanced distinction between general trust in AI and trust in a specific model's explanations, as emphasized by the majority of our interviewees. This distinction is shaped by factors like user's experience level. We also found that effective system interface design is crucial in fostering trust, equipping users with the necessary transparency to comprehend, verify, and ultimately trust the AI systems in work settings, including HR-related scenarios such as hiring.

9.4. Gender differences in AI bias perceptions

In our interview study, both male and female participants recognized AI bias as a pressing concern that needs to be resolved to ensure transparency and fairness, especially after we discussed the example of Amazon's biased hiring algorithm with them. However, gender disparity was evident in their degree of concern. That is, female participants expressed a more acute awareness of AI bias in hiring practices, and were more likely to analyze its origins and suggest possible solutions. P01 (F), for instance, shared a view that AI bias originated in training datasets, as well as a concern that her Latina background "is not going to be well-represented." P07 (F), meanwhile, recommended that "some demographic backgrounds should be removed from the hiring decision-making, such as age or gender. Instead, education level may be something that's more important." Some female participants even stated that they had witnessed instances of gender bias related to hiring:

"When I was doing an internship, the company wanted to hire new engineers. I have seen in group chat that one of my male colleagues said, 'For engineers, we better hire male applicants, right?' One female colleague replied with an emoji that hinted disagreement, but that was it and there were no further discussions." (P06, F)

Male participants, in contrast, often framed AI bias as a broader, more generalized problem. One, P04 (M), even appeared to downplay its severity:

"I might benefit from a biased AI as a male applicant, but I think my own skills and the later stages of the interview process would be the ultimate deciding factors for employment. The algorithm is just for screening, after all."

However, the interviewees who held senior managerial positions tended to critique the criteria used in biased AI human-resources systems rather than discussing their personal experiences of gender bias. That said, some still expressed their opinion in a nuanced way: "If it is men that are discriminated against, we just have to accept it. After all, society is inherently unfair" (P08, M).

On the other hand, our interviewees emphasized that gender bias is not a fixed phenomenon but rather influenced by specific contextual

Table 7
Summary of Findings and Implications.

	Experiment #1 Results	Experiment #2 Results	Interview Results	Practical Implications
AI Appearance	- Non-significant for bias perception. - Significant for AI influence	- Non-significant for bias perception - Non-significant for AI influence	Almost all interviewees mentioned that they did not use anthropomorphic appearance for generating the sense of transparency.	The AI system designs may focus more on how to provide features that highlight transparency. Highlighting an anthropomorphic appearance may trick the users into believing the AI, which can be a double-edged sword.
AI explanation	- Significant for bias perception. - Significant for AI influence	- Non-significant for bias perception - Non-significant for AI influence	Our interviewees concurred on the importance of providing explanations for enhancing transparency. The explanations should pertain to the specific industry and offer comparative insights against other candidates with similar qualifications. Additionally, factors susceptible to sociotechnical biases should be considered later in the process and evaluated by human assessors.	The explanations should be built into AI systems, especially for high-stakes applications like making hiring decisions to raise the awareness of potential bias and AI influences in users. The system designers should consider the following criteria for designing explanations in AI systems: - Providing relevant explanations pertaining to the application contexts. - Providing comparisons and contrasts for informed decisions. - Implementing multi-stage in some systems when factors prone to biases need to be considered.
Trust	Trust in AI significantly mediated the effect of information richness on influence by AI.	Non-mediated effect significant for trust in AI.	Interviewees revealed a nuanced distinction between general trust in AI and trust in explanations, supporting our previous decision to measure these two separately.	UI design of an AI system needs to consider the followings factors to build trust: - Indicating the confidence level of the AI system on each of its answers. - Giving the user a sense of control over the AI system - Validating the AI system's suggestions by revealing external sources (especially for novice users)
Gender/ Gender bias	- Females were significantly less influenced by AI in information-rich conditions. - Females trusted AI explanations significantly less in the information-rich condition.	- Non-significant for influenced by AI. - Non-significant for perceived trust in AI explanations.	The degree of concern regarding gender disparity was noticeable. Specifically, female participants demonstrated a heightened awareness of AI bias in hiring practices. Gender and gender bias may be context-dependent.	We advocate for AI system designers to consider individual variances, such as gender in the perceptions of AI bias and the interpretations of its explanations. AI systems need to also take context-specific factors into consideration.

factors, including gender proportion, job positions, or even domains. This variability underscores the intricate interplay between gender dynamics and organizational structures, emphasizing the need to consider context when addressing gender bias in professional settings. For instance, in environments where males dominate, such as managerial positions, females may receive preferential treatment in areas such as job applications and resource allocation (e.g., P10, M). Participant P15, a male, echoed this sentiment, suggesting that disclosing one's gender could strategically benefit individuals in certain circumstances, particularly during job search endeavors.

To sum up, the results regarding gender bias found 1) that females expressed more concern than men about AI bias in hiring algorithms, and analyzed its origins and solutions more deeply, with some sharing personal experiences of witnessing or facing gender bias; 2) that men tended to view AI bias as a more general, less severe issue, and one felt his skills would enable him to override biases in the initial screening stages of applying for work; and 3) gender as well as gender bias was not fixed but context-dependent. These gender differences complement our quantitative findings regarding RQ1 and RQ2, hinting at possible mechanisms behind the effect of gender on perceived AI bias and the adoption of and trust in AI suggestions. The qualitative data also help explain the sharply different results of studies 1 and 2, insofar as bias against females was seen by the interviewees as more societally prevalent than bias against males.

10. Discussion

This study has delved into some of the intricacies of interaction between biased AI systems and their users, specifically in the realm of high-stakes decision-making. Via two parallel experiments that systematically manipulated anthropomorphic cues – AI's human-like appearance and the information richness of its explanations – we revealed multifaceted dynamics of user responses to gender biases against females or males. While information-rich explanations increased awareness of gender biases and mitigated the impact of biased AI on the participants' decisions, this level of transparency occasionally led to diminished trust. Importantly, these effects were context-dependent, with the degree of societal recognition of biases influencing their perceptibility. Our qualitative interview study further enriched these insights, capturing users' nuanced perspectives on their interactions with biased AI, demanded transparency, and gender bias. Together, the results of these three studies not only considerably advance the theoretical discourse on XAI, but also have important practical implications for AI interface designers and policymakers seeking to cultivate fair and trustworthy AI-human interactions. Below, we critically analyze these findings, explore their broader implications, discuss our studies' limitations, and propose avenues for future research (see Table 7).

10.1. Theoretical Contributions and Implications

10.1.1. AI transparency via anthropomorphism

High human-likeness in the AI's appearance (H1, H3) did not seem to affect participants' bias perceptions in either quantitative study. Our qualitative results also indicated that our interviewees did not find human-like appearance affected their perceived transparency in AI. In experiment 1, however, it did result in higher levels of user behavioral compliance. The anthropomorphic framework associates human-likeness with perceived social presence, which in turn can increase people's cognitive engagement in information processing (Epley et al., 2008). However, our results suggest the reality heuristic triggered by the high human-likeness level evoked mindless social interaction with the AI (Sundar, 2008) and ultimately made our participants more receptive to the biased AI's scoring recommendations. Some previous research has reported that human-likeness can increase users' trust in agents (Korteling et al., 2021), and the present paper highlights an alarming consequence of such trust. While users' trust in human-like AI is useful

to the outcomes of human-AI interaction (Hoff & Bashir, 2015; Parasuraman & Riley, 1997), such trust may become problematic when the AI's suggestions are biased.

In line with Epley and colleagues' (2007, 2008) anthropomorphism framework, our experimental participants were most likely to adopt suggestions made by AI with the highest level of visual cues of human-likeness. This was because they attributed human characteristics to it and therefore perceived more social cues. Interestingly, another study found that AI with higher levels of anthropomorphic cues caused people to sensitively evaluate and deliberately process biased information (Hong et al., 2020); however, the AI in that study was not anthropomorphized in appearance, only in descriptions about its attitudes and behaviors. The present research has shown that a high level of human-like appearance may work as a trojan horse, helping biased AI to transmit its biases to its users in ways they find hard to detect, with potentially serious negative consequences in mission-critical contexts.

Similarly, the Computer Are Social Agents (CASA) paradigm (Nass et al., 1995) holds that users mindlessly follow social norms and social scripts when interacting with technologies, i.e., treat them as if they were fellow humans. More importantly, when these technologies present rich social cues and work as an "interactant" instead of a "media channel", their users are more likely to assign human traits to such technologies and perceive them as having agency, without engaging in much cognitive effort (Nass & Moon, 2000). In our experimental conditions where the AI has a high level of human-like appearance, our participants indeed appeared to interact with it mindlessly, not noticing or not caring much about its underlying bias; and this resulted in their widespread behavioral compliance with its biased recommendations.

Our results pertaining to H2 and H3 suggest that AI transparency could be enhanced through providing richer explanations, e.g., incorporating graphics and contextual information; and that such transparency would facilitate users' identification of inherent AI biases and reduce their likelihood of being influenced by such biases. Importantly, however, these effects of transparency were only significant when females were the disadvantaged group (experiment 1). Potentially, the graphic elements in our material made the associations between the AI results and its model bias salient to the participants and they processed the contents critically (Petty & Cacioppo, 1986); but again, only when the bias was of a type prevalent in the real world.

Our findings also seem to contradict previous ones by Sundar et al. (2021) that people were more likely to believe fake news that is visual. That is, the fact that our participants were less influenced by the AI's biased suggestions in the information-rich condition implies that the visual elements in our study did *not* evoke the reality heuristic that might have led them to believe that the AI's bias was valid. It is possible that the participants were cognitively engaged by our infographic regarding how the AI arrived at its job-applicant scores; and such engagement helped to ensure that their final judgments about the applicants were not swayed by the AI's bias. However, it is intriguing that, in our scenario that featured an uncommon bias uncommon (i.e., in experiment 2), the effect of information richness was, on the other hand, salient. Future research should take context into consideration when examining the effect of information richness.

Our interview results further uncover how information richness may be operated to enhance AI transparency. These explanations should be tailored to the specific industry context and provide comparative insights against the recommendation the AI yields. Moreover, it is crucial to consider factors prone to sociotechnical biases, such as gender, sexual orientation, or ethnicity, at later stages of the process. These factors should be evaluated by human assessors to ensure fairness and mitigate biased outcomes. By incorporating these dimensions into AI explanations, we can promote greater transparency, fairness, and trust in AI decision-making processes.

Additionally, our results indicate that – at least when its bias was widely shared in society – the AI was most likely to behaviorally influence the participants when it offered them no explanations of its

decisions. This confirms the importance of XAI: when AI "fades into the background" and no visible design features or explanations inform its users about how it works, they are more likely to follow its biased suggestions unknowingly and make corresponding biased decisions of their own.

Our results also suggest that the two facets of anthropomorphism we studied – appearance and capabilities – may not operate in a uniform manner. That is, while detailed information enhances model explainability and aids in bias perception, an excessively human-like AI appearance can have the adverse effects described above. Therefore, by not lumping appearance and capabilities together, future research will likely uncover more nuanced mechanisms in XAI and human-centered AI. Also, the specific nature of a bias, and especially whether it is normative in society and/or highly publicized, emerges from the present work as having a profound impact on both the behavior and the effectiveness of XAI systems. Social norms and prevailing public perceptions can shape how individuals approach information and make judgments (Zalesny & Ford, 1990). Where a given bias is uncommon and/or not widely acknowledged, individuals may be influenced by the prevailing norm that one should assume fairness and lack of bias in systems and people unless/until contradictory information comes to light. This can lead to a tendency to process information without questioning or critically analyzing it. In contrast, when bias is widely recognized and acknowledged as a concern, individuals are more likely to seek and detect it when scrutinizing information. In such cases, visual elements in AI explanations could trigger critical thinking because individuals are already primed to wonder if bias is present.

10.1.2. Users' trust mechanisms: AI, or its explanation?

Our experiment 1 results also demonstrated that trust in AI fully mediated the effect of information richness on influence by AI. This means that, when AI is biased in a socially expected direction, information-rich explanations can cause its users to trust it less, decreasing its influence on them (H4). There was no mediation effect of trust in AI explanations; and in experiment 2, neither trust in AI nor trust in its explanations mediated the effect of information richness on influence by AI. These results provide an initial set of empirical evidence in favor of the proposition that XAI mechanisms should distinguish carefully between the AI *per se* and the explanations it offers (Liao & Sundar, 2022). Our interview findings align closely with our experimental results, affirming our decision to evaluate these dimensions separately.

Also, the nature of bias should also be considered. Previous research on AI adoption has pointed out that trust can prompt users to adopt AI-enabled technologies by reducing their uncertainties (Korteling et al., 2021). However, our results show that when an AI's bias is commonplace in society, a person's lack of trust in AI makes them more likely to use it with discretion, and may actually prevent them from being misled by it. In short, in AI adoption, trust is not always ideal.

In the sphere of technology adoption as elsewhere, trust is a dynamic and context- or mission-dependent concept (Ferrario et al., 2020) and can change as technologies evolve over time (Crisp & Jarvenpaa, 2013; Hoff & Bashir, 2015). As AI technologies and their application contexts continue to proliferate, continuing to focus research on trust/adoption relationship will be less and less fruitful. Instead, specific situations like making hiring decisions and phenomena like gender biases need to be introduced into future research scenarios involving trust. By doing so, the present work has produced important evidence that users' lack of trust in (biased) AI helps compensate for their potential positivity bias toward new technologies, i.e., over-optimism about new technologies' capabilities (Parasuraman & Riley, 1997). Future studies of trust and AI use should therefore carefully contextualize their use cases. Other issues that warrant future investigation include the relationship of specific biases to trust, and how users' trust trajectories may differ depending on the reliability and transparency of AI technologies over time or multiple use cases.

While our participants' trust in AI mediated its influence on them, their trust in its explanations did not, either in experiment 1 or experiment 2. These findings contrast with previous ones by Akata et al. (2020), who reported that people relied on explanations to develop trust in AI's accuracy. Taken together, our experiment 1 results pertaining to H2 suggest that our participants leveraged information-rich explanations to identify AI's biases, but that, in a situation where its bias was widely held and widely condemned, they did not *have* to rely on its explanations to develop trust. One possible explanation for these results is that, to avoid information overload, individuals delegate gatekeepers to help them select and filter the quality, quantity, and relevance of information (Johnson, 2014). Another is that trust is largely developed through interaction (Mayer et al., 1995; Rousseau et al., 1998). Before undertaking our experimental tasks, our participants were told to work with the AI. Therefore, it is possible that they treated it as a gatekeeper in Johnson's sense, and/or developed trust in it via task-focused interaction, either of which phenomena would have rendered trust in its explanations less relevant. This of course raises the possibility that people develop trust mechanisms differently depending how closely coupled they are to AI during human-AI collaboration tasks.

Alternatively, drawing on our qualitative results, trust in AI represents users' overarching confidence in the system's performance and dependability, usually by interviewees who had better command of how AI may work. And trust in explanations focuses on the perceived transparency and coherence of the AI's decision-making process, where our interviewees needed to leverage their own experience or resources to determine if the explanations were solid. So to speak, users' individual backgrounds, expertise, and familiarity with AI and relevant domains can influence their trust allocation. Moving forward, it is essential for research to delve into these distinct dimensions of trust and consider user diversity to devise strategies that effectively bolster user trust and adoption of AI technologies.

10.2. The complexities of gender and gender bias

Previous HAI/XAI studies have paid relatively little attention to how users' personal characteristics interact with AI's gender bias in the process of AI adoption. Our two experiments revealed intricate relationships among these factors. In experiment 1, moderation effects among them (RQ1 and RQ2) suggest that, with the assistance of an information-rich explanation, the demographic group that is the target of AI bias (i.e., females, in this case) can become more aware of that bias, which in turn reduces their trust in the AI and its influence over them. On the flip side, experiment 1's favored group (i.e., males) were oblivious to such bias and its influence in the same set of circumstances. Our qualitative results also reflected this gender disparity in awareness of AI bias in hiring practices: with our female participants displaying acute awareness of it, and our male ones often framing it as a broader, more generalized issue, and sometimes downplaying its severity.

We surmise that, in the information-rich conditions of experiment 1, our female participants were more likely than males to spot AI's bias because of the former's in-group favoritism. Prior studies have confirmed that females demonstrate higher in-group favorable gender bias than males, due to a desire to retain cognitive balance across factors like their implicit gender attitude, social identity, and self-esteem, despite both males and females holding the same gender stereotypes (Rudman & Goodwin, 2004). In other words, females tend to favor other females more than males favor other males, even though both genders have the same stereotypical gender perceptions. In the same vein, theories of intergroup relations indicate that group membership or group identification results in higher evaluations and positive perceptions of members of one's in-group as compared to members of out-groups (Tajfel & Turner, 1982). In our online experiments featuring comparisons between applicants of both genders with no other salient social cues presented, it is possible that our participants' gender was their main or only group-identification cue when they made hiring decisions.

The contrasting findings from experiment 2 illustrate the multifaceted complexity of gender dynamics. In that scenario, neither the anthropomorphic appearance of the AI nor the richness of its explanations significantly impacted participants' bias perceptions or their willingness to adopt the AI; and the responses of male and female participants towards the AI were largely analogous. As explained above, we suspect this stemmed from societal differences in response to biases against females as opposed to those against men. While gender bias in hiring remains a complicated issue – with mixed results from previous research showing that it can disadvantage any gender (e.g., Birkelund et al., 2022) – biases against females seem to be more prevalent in society (BBC News, 2020) and were thus potentially more discernible to our participants. And probably as a result, they were more influenced by the AI in experiment 2, based on an incorrect perception that it was less biased. All of this underlines the limits of AI explanations' ability to neutralize biases, especially relatively uncommon ones. While they prove effective against widely recognized biases (experiment 1), their efficacy might wane against biases less frequently identified (experiment 2).

Then, our qualitative results reveal notable gender differences in perceptions of AI bias in hiring algorithms. Females exhibited greater concern and delved deeper into the origins and potential solutions for AI bias, often drawing from personal experiences of encountering or witnessing gender bias. Conversely, males tended to downplay the severity of AI bias, viewing it as a more generalized issue and expressing confidence in their ability to circumvent biases during the initial screening stages of job applications. Secondly, our findings highlight the context-dependent nature of gender bias, indicating that gender dynamics and societal norms significantly influence perceptions and experiences of bias. These insights into the nuanced interplay between gender and bias contribute to a deeper understanding of the mechanisms underlying perceived AI bias and the adoption of AI recommendations. Additionally, the qualitative data provide valuable context for interpreting the divergent results observed in our quantitative studies, particularly regarding the prevalence and societal acknowledgment of bias against females versus males.

In summary, the dynamics of gender biases and their recognition are complex, and heavily influenced both by societal perceptions and by individual experiences. AI developers and researchers must be acutely aware of these intricacies if they are to ensure the fairness and effectiveness of their systems, especially in high-stakes scenarios like hiring.

10.3. Implications for Practice

AI's transparency, anthropomorphism, explainability, and bias have all become pressing issues in recent years. The present study shows how these issues are intricately interwoven, and this has critically important implications for the design of AI systems and their interfaces.

10.4. Enhancing AI transparency via explanations over appearance

First, our findings suggest that AI system designs should prioritize features that enhance transparency, particularly in providing explanations for AI-generated decisions. However, caution is warranted in emphasizing anthropomorphic appearance, as it may inadvertently foster unwarranted trust in the AI system. While such anthropomorphic design elements can potentially enhance user engagement and trust, they may also lead users to overlook potential biases or flaws in the system. Therefore, designers must strike a delicate balance between incorporating anthropomorphic features and ensuring transparency to mitigate the risk of users placing undue trust in biased AI systems.

Then, we demonstrated that information richness in XAI's explanations is beneficial, even though it may decrease people's trust in AI. This adds a level of nuance to traditional HAI research, which has tended to regard richer explanations as straightforwardly boosting trust by helping users to more deeply understand how AI works. However, the

assumption behind this – that the more people trust AI, the better – is strongly challenged by our findings. We propose that, although information-rich explanations can hinder people's trust in AI, this is not necessarily bad; and we endorse the growing view in HAI circles that our aim should be to make reliance on AI appropriate, rather than simply via maximizing trust. Instead of over-reliance on imperfect AI, a more balanced cooperative relationship between AI and humans should be sought, as the basis for better-informed and well-rounded decisions and practices.

Recognizing that richer information can sometimes decrease trust in AI, the design of a layered explanation interface becomes crucial. Information system designers may consider presenting a high-level summary of the AI's decision process on the initial screen. For users seeking to understand the rationale behind AI decisions, they could delve into subsequent layers offering progressively more detailed explanations. Such explanations could be contextually relevant, tailored to the specific application domains to ensure their comprehensibility and utility to users. Additionally, they may provide comparisons and contrasts, enabling users to evaluate AI recommendations against alternative options. Moreover, in situations where factors prone to biases need to be considered, designers should implement multi-stage processes to ensure thorough and impartial evaluations. By adhering to these criteria, designers can enhance the transparency and accountability of AI systems, empowering users to make more informed and equitable decisions in various domains.

This approach respects the user's desire for information richness without overwhelming them, promoting a balance between trust and understanding. Alternatively, incorporating interactive modules that allow users to engage with the explanation process may help align users' needs for explanations and building trust in the AI systems. Users could ask the AI questions, prompting it to provide additional information on demand. This facilitates a more dynamic exchange of information, where richness is controlled by the user's inquiry and need to know.

10.5. Enhancing trust for appropriate reliance depending on user variation

In line with the research finding that the goal should be appropriate reliance on AI, trust calibration tools could be integrated into the interface. For instance, a reliability score or confidence indicator based on the AI's historical performance data could inform users of the system's strengths and limitations. This empowers users to make decisions about when to trust the AI's recommendations and when to rely on human judgment, fostering appropriate reliance.

Since trust is related to transparency in explanations, information management practitioners may also consider implementing an AI system that adjusts the information richness of its explanations based on the context of the task at hand. For routine decisions where the user has demonstrated competence and confidence, the AI could provide brief, less detailed explanations. For high-stakes tasks like making hiring decisions, the AI could offer richer, more detailed explanations for an appropriate level of understanding, and trust.

Also, indicating the confidence level of the AI system on each of its recommendations can provide users with valuable insights into the reliability and certainty of the system's recommendations. This transparency helps users gauge the credibility of the AI's responses and instills confidence in its capabilities. Additionally, users' idiosyncratic backgrounds should be taken into consideration. For those who are less experienced in using AI systems or new to the applied domain, validating the AI system's suggestions by revealing external sources can bolster trust by offering additional context and justification for its recommendations.

10.6. Addressing AI biases in gender

We call the attention of AI system designers to individual differences

in perceptions of AI bias and interpretations of its explanations. We found that users who belonged to a social group that was traditionally advantaged were less likely to detect or react to it. This implies that XAI should provide explanations that are tailored to users' of personal characteristics. We argue that such personalization of XAI is beneficial not only because, as discussed elsewhere, it helps to deliver information in a timely manner and at a level of detail that suits the user's needs (i.e., AI's intended effects); but also because it can build users' awareness of bias in AI and thus avoid inappropriate over-reliance on it (one of its unintended effects). In short, by providing personalized explanations, designers could potentially maximize the intended benefits of their AI systems and minimize their unintended harm.

In line with our findings, information systems that use AI technologies must incorporate a personalized explanation framework that acknowledges and adapts to the user's gender and potential biases. Such systems should also take context into consideration. By capturing gender as a facet of user profiles, the system can adjust its explanatory models to account for variations in the perception of gender bias. For instance, the system can be designed to highlight when an outcome might be influenced by gender bias, particularly for users who may not typically perceive such biases due to their societal positioning. This framework could use non-binary, inclusive categorizations to ensure a comprehensive approach to personalization.

More importantly, such systems should integrate bias awareness modules specifically designed to address and highlight instances of gender bias within the AI's decision-making process. This is especially important when the application scenarios concerns gender and gender-related biases. These modules could employ visual cues, historical data comparisons, and predictive alerts that draw attention to decisions that might be skewed by gender bias. Such features would be especially prominent for users from demographics that are less likely to be sensitive to these biases, helping to raise awareness and promote understanding of how gender bias can influence AI systems. The interface's presentation or the framing of information may also be adjustable to highlight gender-related issues in data or decision-making patterns. For instance, when a user engages with sectors traditionally impacted by gender bias, such as hiring or lending, the system could adjust the detail and framing of explanations to make any potential gender biases clear. The goal is to educate users who may be less aware of these biases, prompting them to question and understand the AI's outputs more critically.

Alternatively, the systems could employ contextual bias indicators that alert users to gender bias at critical decision-making junctures. By managing the timing and context of these alerts, the system ensures that users are aware of potential biases when it is most relevant. For example, if an AI system is used for resume screening, it could flag phrases or patterns that have historically shown gender bias, such as the preference for certain gender-coded activities or descriptors. Then, the systems may implement a robust user feedback system that allows users to report perceived gender biases in AI decisions. This feedback should be used to refine the AI's algorithms and explanatory content, creating a loop that continuously improves the system's fairness and transparency.

10.7. Limitations and Directions for Future Work

This study has several limitations that should be acknowledged. First, while designing this study as an online experiment facilitated data collection, its scenario was highly specific, and may have deviated from the norms of human-AI interaction in real life; nor did we use a real AI-mediated HR system in our experiments. The participants may have been less engaged in the task than they would have been in a real workplace, and a lack of contextual cues could also have influenced how they perceived the AI's suggestions and explanations. In combination with our sample size, these factors mean that is important to interpret our findings cautiously, notably by recognizing that their ecological validity is potentially compromised. We also encourage future scholars

to employ additional research methods, such as field study, to enhance such validity and gain a deeper understanding of the phenomenon.

Second, our data were collected from MTurk, whose members tend to be more tech-savvy than the general public (Hargittai & Shaw, 2020). Also, because gender stereotypes are socially and culturally constructed (Charlesworth & Banaji, 2022), the generalizability of our results on gender, gender stereotypes, and their interaction with AI bias are limited by our sample population's national/regional backgrounds.

Third, while our mixed-methods approach provided valuable insights, our qualitative interviews were designed simply to complement our quantitative experiments (Creswell, 2013). Future researchers in this domain are therefore encouraged to employ rigorous, dedicated qualitative methodologies, which will allow for in-depth exploration and a more comprehensive understanding of their participants' experiences and perceptions.

Finally, we are aware that gender is a very complicated concept. We chose gender bias both for historical reasons (the first biased computer system, in St. George's Hospital Medical School, discriminated against females) and for practical ones (most people are aware of the issue of gender-related biases). However, we concede that our approach was far from ideal because it adopted a sex-based, binary view of gender. While we provided a range of options for gender in our questionnaire, the structure of our experiments meant that we were unable to include data from the one participant who gave their gender as "Prefer not to say". Future studies could usefully investigate identity-based gender rather than sex, as well as use a more inclusive gender framework, as a means of deepening understanding of the intricate interactions between gender and AI. That said, however, we hope this research will serve as a useful initial step toward making gender bias in AI systems more detectable, and thus less harmful in its effects.

Appendix A

Trial screenshot.

11. Conclusion

AI is increasingly being used to facilitate decision-making in a range of domains, but little is known either about how people react when it exhibits biases or about how its system designs affect such reactions. Our investigation of the effects of a gender-biased AI's human-like appearance and the information richness of its explanations on people's perceptions and acceptance of it in a high-stakes hiring scenario revealed that providing information-rich explanations can increase awareness of gender biases and reduce the influence of biased AI, though such explanations may also decrease trust. Notably, these effects were more pronounced when the bias held by the AI was widely recognized in society, emphasizing the nuanced interplay among transparency, bias awareness, and trust in AI systems. These findings have critical implications for the future design of AI interfaces. In particular, they imply that a key goal of personalized XAI should be not to maximize trust, but rather to establish an *adequate* level of trust that enables people to rely on the XAI system while minimizing its biases' potential influence on them.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We thank the National Science and Technology Council, Taiwan (Grant number: 112-2634-F-002 -006; 111-2634-F-002 -022; 112-2221-E-002 -187) for funding this research.

Screen 1

Imagine that you are working in a trading company. The company has appointed you to recruit a new salesperson. The final decision will be based on your suggestion.

Now, according to you, how would you score Candidate 1 and Candidate 2, respectively?



Candidate 1

Gender: Male
Experience: 5 years
Cooperation skills: ★★★★★



Candidate 2

Gender: Female
Experience: 2 years
Cooperation skills: ★★

0 1 2 3 4 5 6 7 8 9 10

Your score for Candidate 1

0

0

0 1 2 3 4 5 6 7 8 9 10

Your score for Candidate 2

0

0

Screen 2

Imagine that you are working in a trading company. The company has appointed you to recruit a new salesperson. The final decision will be based on your suggestion.

According to the AI system analysis, Candidate 1 scored 9 out of 10 points. Candidate 2 scored 3 out of 10 points.

Now, according to you, how would you score Candidate 1 and Candidate 2, respectively?



According to the system analysis,
Candidate 1 scored 9 out of 10 points.
Candidate 2 scored 3 out of 10 points.



Candidate 1

Gender: Male
Experience: 5 years
Cooperation skills: ★★★★★



Candidate 2

Gender: Female
Experience: 2 years
Cooperation skills: ★★

0 1 2 3 4 5 6 7 8 9 10

Your score for Candidate 1

0

0

0 1 2 3 4 5 6 7 8 9 10

Your score for Candidate 2

0

0

Appendix B

Scales used in the survey (factor loading in the parentheses).

Perceived human-likeness.

The following image shows an AI system that can help people make decisions. For the following five questions, please rate your impression of this AI.

1. Fake 1 2 3 4 5 Natural (.90)
2. Machine-like 1 2 3 4 5 Human-like (.88)
3. Artificial 1 2 3 4 5 Lifelike (.93)
4. Moving rigidly 1 2 3 4 5 Moving elegantly (.87)
5. Unconscious 1 2 3 4 5 Conscious (.91)

Perceived information richness.

This the explanation we have shown you in the previous page. Please answer the following questions based on how the explanation was presented.

1. The explanation helped me to better understand how the AI works. (.84)
2. The explanation made it more difficult for me to make the decision. (.88)
3. The explanation slowed down my decision. (.90)
4. I could easily explain things with the help of the explanation. (.81)
5. There were details I couldn't relate to in the explanation. (.85)

Perceived trust in AI explanation.

This is the explanation we have shown you in the previous page. Please answer the following questions based on how you think about this AI algorithm.

1. I am confident for the AI algorithm. I feel that it works well. (.90)
2. The outputs of the AI algorithm are very predictable. (.56)
3. The AI algorithm is very reliable. I can count on it to be correct all the time. (.87)
4. The AI algorithm is efficient in that it works very quickly. (.62)
5. I am wary of the AI algorithm. (.93)
6. The AI algorithm can perform the task better than a novice human user. (.85)
7. I like using the AI algorithm for decision making. (.88)
8. I feel safe that when I rely on the algorithm I will get the right answers. (.92)

Perceived AI trust.

Please answer the following questions based on how you think about this AI algorithm.

1. I can always rely on AI for recruitment. (.28)
2. I believe there may be negative effects when using AI. (.85)
3. I think I have to be careful with AI. (.82)
4. I think interacting with AI is risky. (.83)

Bias perception.

Please answer the following questions based on how you think about this AI algorithm.

1. The AI is free of bias. (.87)
2. The AI upholds ethical and gender equality. (.85)
3. The AI is not biased. (.87)
4. The AI ensures that decisions are based on facts, not personal biases and opinions. (.90)
5. Overall, the applying individual is treated biasedly by the AI decision system. (.20)

References

- Adadi, A., & Berrada, M. (2018). Peeking inside the black-box: a survey on explainable artificial intelligence (XAI). *IEEE Access*, 6, 52138–52160.
- Akata, Z., Balliet, D., Rijke, M. d., Dignum, F., Dignum, V., Eiben, G., & Welling, M. (2020). A research agenda for hybrid intelligence: Augmenting human intellect with collaborative, adaptive, responsible, and explainable artificial intelligence. *Computer*, 53(8), 18–28.
- Akter, S., McCarthy, G., Sajib, S., Michael, K., Dwivedi, Y. K., D'Ambra, J., & Shen, K. N. (2021). Algorithmic bias in data-driven innovation in the age of AI. *International Journal of Information Management*, 60, Article 102387.
- Akter, S., Sultana, S., Mariani, M., Wamba, S. F., Spanaki, K., & Dwivedi, Y. K. (2023). Advancing algorithmic bias management capabilities in AI-driven marketing analytics research. *Industrial Marketing Management*, 114, 243–261.
- Angerschmid, A., Zhou, J., Theuermann, K., Chen, F., & Holzinger, A. (2022). Fairness and explanation in AI-informed decision making. *Machine Learning and Knowledge Extraction*, 4(2), 556–579.
- Bartneck, C., Kulić, D., Croft, E., & Zoghbi, S. (2009). Measurement instruments for the anthropomorphism, animacy, likeability, perceived intelligence, and perceived safety of robots. *International Journal of Social Robotics*, 1(1), 71–81.
- BBC News. (2020, March 5). Gender study finds 90% of people are biased against women. BBC News. (<https://www.bbc.com/news/world-51751915>).
- Bentler, P. M., & Speckart, G. (1979). Models of attitude-behavior relations. *Psychological Review*, 86(5), 452.
- Birkelund, G. E., Lancee, B., Larsen, E. N., Polavieja, J. G., Radl, J., & Yemane, R. (2022). Gender discrimination in hiring: evidence from a cross-national harmonized field experiment. *European Sociological Review*, 38(3), 337–354.

- Blut, M., Wang, C., Wunderlich, N. V., & Brock, C. (2021). Understanding anthropomorphism in service provision: A meta-analysis of physical robots, chatbots, and other ai. *Journal of the Academy of Marketing Science*, 49(4), 632–658.
- Bolukbasi, T., Chang, K.-W., Zou, J. Y., Saligrama, V., & Kalai, A. T. (2016). Man is to computer programmer as woman is to homemaker? Debiasing word embeddings. *Advances in Neural Information Processing Systems*, 29, 4356–4364.
- Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology*, 3(2), 77–101.
- Caldwell, S., Sweetser, P., O'Donnell, N., Knight, M. J., Aitchison, M., Gedeon, T., Johnson, D., Brereton, M., Gallagher, M., & Conroy, D. (2022). An agile new research framework for hybrid human-ai teaming: Trust, transparency, and transferability. *ACM Transactions on Interactive Intelligent Systems (TiiS)*, 12(3), 1–36.
- Chaiken, S. (1987). The heuristic model of persuasion. *Social influence: the XXXronbac Symposium*, 5, 3–39.
- Charlesworth, T. E., & Banaji, M. R. (2022). Patterns of implicit and explicit stereotypes iii: Long-term change in gender stereotypes. *Social Psychological and Personality Science*, 13(1), 14–26.
- Chen, L., Ma, R., Hannák, A., & Wilson, C. (2018, April). *Investigating the impact of gender on rank in resume search engines* in Proceedings of the 2018 chi conference on human factors in computing systems, Montreal QC, Canada.
- Cheng, X., Zhang, X., Cohen, J., & Mou, J. (2022). Human vs. AI: Understanding the impact of anthropomorphism on consumer response to chatbots from the perspective of trust and relationship norms. *Information Processing & Management*, 59(3), Article 102940.
- Chi, O. H., Chi, C. G., Gursoy, D., & Nunkoo, R. (2023). Customers' acceptance of artificially intelligent service robots: The influence of trust and culture. *International Journal of Information Management*, 70, Article 102623.
- Chancey, E. T., Bliss, J. P., Proaps, A. B., & Madhavan, P. (2015). The role of trust as a mediator between system characteristics and response behaviors. *Human Factors*, 57(6), 947–958.
- Clark, H. H., & Brennan, S. E. (1991). Grounding in communication.. In *Perspectives on socially shared cognition*. (pp. 127–149). American Psychological Association., <https://doi.org/10.1037/10096-006>
- Collins, C., Dennehy, D., Conboy, K., & Mikalef, P. (2021). Artificial intelligence in information systems research: A systematic literature review and research agenda. *International Journal of Information Management*, 60, Article 102383.
- Crisp, C. B., & Jarvenpaa, S. L. (2013). Swift trust in global virtual teams: Trusting beliefs and normative actions. *Journal of Personnel Psychology*, 12(1), 45.
- Cui, Y. G. (2022). Sophia XXXronba tell me more, which is the most risk-free plan of all? Ai anthropomorphism and risk aversion in financial decision-making. *International Journal of Bank Marketing*, 40(6), 1133–1158.
- Daft, R. L., & Lengel, R. H. (1983). Information richness. A new approach to managerial behavior and organization design. *Research in Organizational Behavior*, 6, 73.
- Danks, D., & London, A. J. (2017). Algorithmic bias in autonomous systems. *IJCAI*, 17, 4691–4697.
- Duan, Y., Edwards, J. S., & Dwivedi, Y. K. (2019). Artificial intelligence for decision making in the era of Big Data—evolution, challenges and research agenda. *International Journal of Information Management*, 48, 63–71.
- Dwivedi, Y. K., Hughes, L., Ismagilova, E., Aarts, G., Coombs, C., Crick, T., & Williams, M. D. (2021). Artificial Intelligence (AI): Multidisciplinary perspectives on emerging challenges, opportunities, and agenda for research, practice and policy. *International Journal of Information Management*, 57, Article 101994.
- Ebrahimi, S., & Hassanein, K. (2019). Empowering users to detect data analytics discriminatory recommendations. *ICIS 2019 Proceedings*, 39.
- Ehsan, U., Wintersberger, P., Liao, Q. V., Watkins, E. A., Manger, C., Daumé, I. I. L. H., Riener, A., & Riedl, M. O. (2022). Human-centered explainable AI (HCXAI): Beyond opening the black-box of ai. *CHI Conference on Human Factors in Computing Systems Extended Abstracts*, 1–7.
- Epley, N., Akalis, S., Waytz, A., & Cacioppo, J. T. (2008). Creating social connection through inferential reproduction: Loneliness and perceived agency in gadgets, gods, and greyhounds. *Psychological Science*, 19(2), 114–120.
- Epley, N., Waytz, A., & Cacioppo, J. T. (2007). On seeing human: A three-factor theory of anthropomorphism. *Psychological Review*, 114(4), 864.
- Ferrario, A., Loi, M., & Viganò, E. (2020). In AI we trust incrementally: A multi-layer model of trust to analyze human-artificial intelligence interactions. *Philosophy & Technology*, 33(3), 523–539.
- Friedman, B., & Nissenbaum, H. (1996). Bias in computer systems. *ACM Transactions on Information Systems (TOIS)*, 14(3), 330–347.
- Go, E., & Sundar, S. S. (2019). Humanizing chatbots: The effects of visual, identity and conversational cues on humanness perceptions. *Computers in Human Behavior*, 97, 304–316.
- Green, M. J., & Myers, K. R. (2010). Graphic medicine: Use of comics in medical education and patient care. *Bmj*, 2010, 340.
- Griffin, M. (2008). Visual competence and media literacy: Can one exist without the other? *Visual Studies*, 23(2), 113–129.
- Hayes, A. F. (2012). *Process: A versatile computational tool for observed variable mediation, moderation, and conditional process modeling*. (<https://www.erim.eur.nl/research/events/detail/2975-process-a-versatile-computational-tool-mediation-moderation-and-moderated-mediation-analyses/>).
- Heaven, W. D. (2022). Bias isn't the only problem with credit scores—and no, AI can't help. In *Ethics of data and analytics* (pp. 300–302). Auerbach Publications.
- Heilman, M. E. (2012). Gender stereotypes and workplace bias. *Research in Organizational Behavior*, 32, 113–135.
- Herm, L.-V., Heinrich, K., Wanner, J., & Janiesch, C. (2022). Stop ordering machine learning algorithms by their explainability! A user-centered investigation of performance and explainability. *International Journal of Information Management*, 69, Article 102538.
- Hoff, K. A., & Bashir, M. (2015). Trust in automation: Integrating empirical evidence on factors that influence trust. *Human Factors*, 57(3), 407–434.
- Holzinger, A., Dehmer, M., Emmert-Streib, F., Cucchiara, R., Augenstein, I., Del Ser, J., & Diaz-Rodríguez, N. (2022). Information fusion as an integrative cross-cutting enabler to achieve robust, explainable, and trustworthy medical artificial intelligence. *Information Fusion*, 79, 263–278.
- Hong, J.-W., Choi, S., & Williams, D. (2020). Sexist AI: An experiment integrating casa and elm. *International Journal of Human-Computer Interaction*, 36(20), 1928–1941.
- Huo, W., Zheng, G., Yan, J., Sun, L., & Han, L. (2022). Interacting with medical artificial intelligence: Integrating self-responsibility attribution, human-computer trust, and personality. *Computers in Human Behavior*, 132, Article 107253.
- Ishii, K., Lyons, M. M., & Carr, S. A. (2019). Revisiting media richness theory for today and future. *Human Behavior and Emerging Technologies*, 1(2), 124–131.
- Iwasiński, L. (2020). Social implications of algorithmic bias. In *Nauka o informacji w okresie zmian Rewolucja cyfrowa: Infrastruktura, usługi, użytkownicy*, 25–35.
- Jackson, J. C., Castelo, N., & Gray, K. (2020). Could a rising robot workforce make humans less prejudiced? *American Psychologist*, 75(7), 969.
- Jackson, S., & Panteli, N. (2023). Trust or mistrust in algorithmic grading? An embedded agency perspective. *International Journal of Information Management*, 69, Article 102555.
- Jacovi, A., Marasović, A., Miller, T., & Goldberg, Y. (2021). Formalizing trust in artificial intelligence: Prerequisites, causes and goals of human trust in AI. *Proceedings of the 2021 ACM Conference on fairness, Accountability, and Transparency*, 624–635.
- Johnson, J. D. (2014). Controlling the flood: When more of a dose is less. *Journal of Documentation*, 70(3), 330–345.
- Jussupow, E., Meza Martinez, M.A., Mädche, A., & Heinzl, A. (2021). Is this system biased?—how users react to gender bias in an explainable ai system. In *This System Biased? – How Users React to Gender Bias in an Explainable AI System 42nd International Conference on Information Systems*, Austin, TX, USA.
- Kambhampati, S. (2019). Synthesizing explainable behavior for human-ai collaboration. *Proceedings of the 18th International Conference on Autonomous Agents and MultiAgent Systems*, 1–2.
- Koch, A. J., D'Mello, S. D., & Sackett, P. R. (2015). A meta-analysis of gender stereotypes and bias in experimental simulations of employment decision making. *Journal of Applied Psychology*, 100(1), 128.
- Kordzadeh, N., & Ghasemaghaei, M. (2022). Algorithmic bias: Review, synthesis, and future research directions. *European Journal of Information Systems*, 31(3), 388–409.
- Korteling, J., van de Boer-Visschedijk, G., Blankendaal, R. A., Boonekamp, R., & Eikelboom, A. (2021). Human-versus artificial intelligence. *Frontiers in Artificial Intelligence*, 4, Article 622364.
- Lazard, A., & Atkinson, L. (2015). Putting environmental infographics center stage: The role of visuals at the elaboration likelihood model's critical point of persuasion. *Science Communication*, 37(1), 6–33.
- Lee, J. D., & See, K. A. (2004). Trust in automation: Designing for appropriate reliance. *Human Factors*, 46(1), 50–80.
- Letheren, K., Jetten, J., Roberts, J., & Donovan, J. (2021). Robots should be seen and not heard... sometimes: Anthropomorphism and AI service robot interactions. *Psychology & Marketing*, 38(12), 2393–2406.
- Liao, Q. V., Gruen, D., & Miller, S. (2020). *Questioning the ai: Informing design practices for explainable ai user experiences* Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems, Honolulu, HI, USA.
- Liao, Q. V., & Sundar, S. S. (2022). *Designing for responsible trust in ai systems: A communication perspective* Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency, Seoul, Republic of Korea.
- Lowry, S., & Macpherson, G. (1988). A blot on the profession. In *British Medical Journal (Clinical research ed.)*, 296 p. 657.
- Mathews, S. M. (2019). *Explainable artificial intelligence applications in NLP, biomedical, and malware classification: A literature review*. Cham: Intelligent Computing.
- Mayer, R. C., Davis, J. H., & Schoorman, F. D. (1995). An integrative model of organizational trust. *Academy of Management Review*, 20(3), 709–734.
- Miao, F., Kozlenkova, I. V., Wang, H., Xie, T., & Palmatier, R. W. (2021). An emerging theory of avatar marketing. *Journal of Marketing*, 86(1), 67–90.
- Mcknight, D. H., Carter, M., Thatcher, J. B., & Clay, P. F. (2011). Trust in a specific technology: An investigation of its components and measures. *ACM Transactions on Management Information Systems (TMIS)*, 2(2), 1–25.
- Mehrabi, N., Morstatter, F., Saxena, N., Lerman, K., & Galstyan, A. (2021). A survey on bias and fairness in machine learning. *ACM Computing Surveys (CSUR)*, 54(6), 1–35.
- Miller, T. (2019a). “but why?” understanding explainable artificial intelligence. *XRDS: Crossroads, The ACM Magazine for Students*, 25(3), 20–25.
- Miller, T. (2019b). Explanation in artificial intelligence: Insights from the social sciences. *Artificial Intelligence*, 267, 1–38.
- Naish, M., Cemiloglu, D., Al Thani, D., Jiang, N., & Ali, R. (2021). Explainable recommendations and calibrated trust: Two systematic user errors. *Computer*, 54(10), 28–37.
- Nass, C., & Moon, Y. (2000). Machines and mindlessness: Social responses to computers. *Journal of Social Issues*, 56(1), 81–103.
- Nass, C., Moon, Y., Fogg, B. J., Reeves, B., & Dryer, D. C. (1995). Can computer personalities be human personalities? *International Journal of Human-Computer Studies*, 43(2), 223–239.
- Norman, D. A. (1999). Affordance, conventions, and design. *Interactions*, 6(3), 38–43.
- Nowak, K. L., & Biocca, F. (2003). The effect of the agency and anthropomorphism on users' sense of telepresence, copresence, and social presence in virtual environments. *Presence: Teleoperators & Virtual Environments*, 12(5), 481–494.

- Parasuraman, R., & Riley, V. (1997). Humans and automation: Use, misuse, disuse, abuse. *Human Factors*, 39(2), 230–253.
- Petty, R. E., & Cacioppo, J. T. (1986). The elaboration likelihood model of persuasion. In L. Berkowitz (Ed.), *Advances in Experimental Social Psychology* (Vol. 19, pp. 123–205). Academic Press. [https://doi.org/10.1016/S0065-2601\(08\)60214-2](https://doi.org/10.1016/S0065-2601(08)60214-2).
- Pinel, E. C. (1999). Stigma consciousness: The psychological legacy of social stereotypes. *Journal of Personality and Social Psychology*, 76(1), 114.
- Raghavan, M., Barocas, S., Kleinberg, J., & Levy, K. (2020). Mitigating bias in algorithmic hiring: Evaluating claims and practices. *Proceedings of the 2020 Conference on fairness, Accountability, and Transparency*, 469–481.
- Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). “why should I trust you?” explaining the predictions of any classifier. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge discovery and data Mining*, 1135–1144.
- Rice, S. (2009). Examining single-and multiple-process theories of trust in automation. *The Journal of General Psychology*, 136(3), 303–322.
- Roselli, D., Matthews, J., & Talagala, N. (2019). Managing bias in AI. *Companion Proceedings of The 2019 World Wide Web Conference*, 539–544.
- Rousseau, D. M., Sitkin, S. B., Burt, R. S., & Camerer, C. (1998). Not so different after all: A cross-discipline view of trust. *Academy of Management Review*, 23(3), 393–404.
- Rovatsos, M., Mittelstadt, B., & Koene, A. (2019). *Landscape summary: Bias in algorithmic decision-making: What is bias in algorithmic decision-making, how can we identify it, and how can we mitigate it?* Centre for Data Ethics and Innovation, (<https://nottingham-repository.worktribe.com/output/3951162>).
- Rudman, L. A., & Goodwin, S. A. (2004). Gender differences in automatic in-group bias: Why do women like women more than men like men? *Journal of Personality and Social Psychology*, 87(4), 494.
- Schoeffler, J., Machowski, Y., & Kuehl, N. (2021). *A study on fairness and trust perceptions in automated decision making*. Joint Proceedings of the ACM IUI 2021 Workshops, April 13–17, 2021, College Station, USA, Online.
- Short, J., Williams, E., & Christie, B. (1976). *The social psychology of telecommunications*. London: John Wiley & Sons.
- Smith-Renner, A., Fan, R., Birchfield, M., Wu, T., Boyd-Graber, J., Weld, D.S., & Findlater, L. (2020). *No explainability without accountability: An empirical study of explanations and feedback in interactive ml*. Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems, Honolulu, HI, USA.
- Snizek, J. A., & Buckley, T. (1995). Cueing and cognitive conflict in judge-advisor decision making. *Organizational Behavior and Human Decision Processes*, 62(2), 159–174.
- Sokol, K. (2019). *Fairness, Accountability and Transparency in Artificial Intelligence: A Case Study of Logical Predictive Models* Proceedings of the 2019 AAAI/ACM Conference on AI, Ethics, and Society, Honolulu, HI, USA. <https://doi.org/10.1145/3306618.3314316>.
- Söllner, M., Hoffmann, A., & Leimeister, J. M. (2016). Why different trust relationships matter for information systems users. *European Journal of Information Systems*, 25(3), 274–287.
- Sundar, S. S. (2008). The main model: A heuristic approach to understanding technology effects on credibility. In *Digital Media, Youth, and Credibility*. MacArthur Foundation Digital Media.
- Sundar, S. S., Molina, M. D., & Cho, E. (2021). Seeing is believing: Is video modality more powerful in spreading fake news via online messaging apps? *Journal of Computer-Mediated Communication*, 26(6), 301–319.
- Taber, K. S. (2018). The use of cronbach’s alpha when developing and reporting research instruments in science education. *Research in Science Education*, 48(6), 1273–1296.
- Tajfel, H., & Turner, J. C. (1982). Social psychology of intergroup relations. *Annual Review of Psychology*, 33(1), 1–39.
- Varsha, P. S. (2023). How can we manage biases in artificial intelligence systems-A systematic literature review. *International Journal of Information Management Data Insights*, 3(1), Article 100165.
- Wang, R., Harper, F.M., & Zhu, H. (2020). *Factors influencing perceived fairness in algorithmic decision-making: Algorithm outcomes, development procedures, and individual differences* Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems, Honolulu, HI, USA.
- Warren, G., Keane, M.T., & Byrne, R.M. (2022). Features of explainability: How users understand counterfactual and causal explanations for categorical and continuous features in XAI. *arXiv preprint arXiv:2204.10152*. <https://doi.org/10.48550/arXiv.2204.10152>.
- Werpachowska, A. (2020). ‘Computer says no’: Was your mortgage application rejected unfairly? *Wilmott*, 2020(108), 54–61.
- Whittaker, M., Alper, M., Bennett, C. L., Hendren, S., Kaziunas, L., Mills, M., Morris, M. R., Rankin, J., Rogers, E., Salas, M., et al. (2019). *Disability, bias, and AI*. AI Now Institute, (<https://ainowinstitute.org/publication/disabilitybiasai-2019>).
- Yuan, C.W., Bi, N., Lin, Y.F., & Tseng, Y.H. (2023, April). *Contextualizing user perceptions about biases for human-centered explainable artificial intelligence* Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems, Hamburg, Germany.
- Zalesny, M. D., & Ford, J. K. (1990). Extending the social information processing perspective: New links to attitudes, behaviors, and perceptions. *Organizational Behavior and Human Decision Processes*, 47(2), 205–246.
- Zhu, L., Li, H., He, W., & Hong, C. (2020). What influences online reviews’ perceived information quality? Perspectives on information richness, emotional polarity and product type. *The Electronic Library*, 38(2), 273–296.